



RX INTERVIEWS

**BONUS 2023-4 INTERVIEW
QUESTIONS**

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INTRODUCTION

In keeping with my annual tradition, this 2023-24 Bonus Guide will cover a set of more obscure or difficult questions – or those that aren't too hard but that are worth covering for the sake of completeness – that came up over the last recruiting cycle (**almost all of these were for full-time or lateral interviews**). Plus, I've added a bunch of my own questions to highlight certain concepts that I think are worth talking about a bit more.

However, it's important to remember that everything covered here should be treated as additive to, and not a substitute for, the material in the Overview Guide, Bond Math Guide, Structural Subordination Guide, Bonus 2021 Guide, Bonus 2022 Guide, etc. It all builds on each other. For example, even though a common question last cycle involved a waterfall where there's a deficiency claim that kind of question isn't covered here as it was covered in the Bonus 2022 Guide – but that doesn't mean it's not liable to come up in future interviews (it will!).

Like I also mentioned at the outset of the Bonus 2022 Guide, many of the questions below will be a bit trickier than the standard questions that'll be asked, and my explanations (as per usual) will often be much more in-depth than you'd be expected to deliver. So, if you read some of the questions below and start to feel a bit lost in the sauce, don't worry about it too much. **Since I've covered so much in all the other guides, these questions somewhat represent the dregs of restructuring interviews (read: the kinds of questions that are rare to come up)**. Given this, if you're short on prep time and long on interview-related anxiety, it's probably a better use of time to focus on the Bonus 2021 Guide and the Bonus 2022 Guide as those contain more common and standard questions – then you can circle back here time permitting.

Further, all of the interview questions that I created are more or less an excuse for me to explain certain concepts a bit further and, as you'd expect, tend to tilt to the more difficult side (although, as a fair warning, the questions I create have a funny way of ending up being interview questions in future cycles...).

So, with that disclaimer out of the way, let's get started...

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- 1 **Let's say there's a company with a 20x P/E, a 10x EV/EBITDA, \$20mm of annual interest expense, \$20mm of annual D&A, an interest rate of 5% on its debt, and a \$200mm market cap. Given all this, what's the company's tax rate?**

Let's start off with a nice non-rx question here. Whenever you get a question with a bunch of building blocks it can sometimes be a bit difficult to figure out where to begin. But instead of sitting back and hoping for the right approach to reveal itself it's never a bad idea to start off trying to figure out what you can build in one or two steps and seeing where that leads you.

For example, we've been given the annual interest expense and the interest rate. And we know that the former divided by the latter will give us the overall level of debt ($20\text{mm} / 0.05 = 400\text{mm}$). Likewise, we know that EV, with the limited info we have, will just be $\text{EV} = \text{Market Cap} + \text{Debt}$, and we now have both the market cap and the total debt. So, EV is \$600mm and, since we know $\text{EV} / \text{EBITDA} = 10\text{x}$, then that means EBITDA is \$60mm.



Now, we know that the P/E ratio is (Price Per Share / Earnings Per Share). However, we haven't been given the number of shares outstanding. Instead, we've been given the market cap. But we know that's simply the price per share multiplied by the number of shares outstanding. So, another way to express P/E – removing the share count consideration, although you'd get the same answer if you assumed some arbitrary number of shares outstanding – is (Market Cap / Earnings) or $(20x = 200\text{mm} / \text{Earnings})$ or $200\text{mm} / 20 = 10\text{mm}$. Therefore, we know that our after-tax earnings are \$10mm and that if we find the pre-tax earnings we'll then be able to find our tax rate.

So, to that end, we know that our interest expense is \$20mm and the level of annual D&A is \$20mm. Therefore, since we found earlier that we have EBITDA of \$60mm after deducting interest expense and D&A our *pre-tax* earnings will be \$20mm and, since we now know our *after-tax* earnings are \$10mm, that means we must have \$10mm in taxes or a 50% tax rate.

1 Let's say we're looking at some Senior Secured Notes with a coupon of 8.0% and a Term Loan at S + 700bps with a SOFR floor of 100bps. Both have the same maturity and reside pari to each other. Further, let's assume that SOFR is 50bps right now. Given all this, if we assume both are trading at the same price which would you prefer to invest in?

There are a number of things that we could consider here (e.g., structural considerations, if the TL benefits from grid pricing, etc.). However, the purpose of this question and what the interviewer will want you to recognize is that at a minimum these two pari pieces of debt are going to earn you the same amount (8%) due to the TL benefiting from a floor. *But* in a rising rates environment, as we've seen over the past two years, the SSNs will have us clipping the same amount of coupons (since the SSNs are fixed-rate) while the TL interest will rise with SOFR (e.g., today, since SOFR is a little over 5%, we'd be looking at earning over 12% on 1L paper here).

Note: This is why Steve Schwarzman said his [hilarious \(maybe sad if contemplated too much?\) line](#) vis-à-vis the growth of private credit: "If you can earn 12%, maybe 13% on a really good day in senior secured bank debt, what else do you want to do in life? If you are living in a no-growth economy and somebody can give you 12-13% with almost no prospect of loss, that's about the best thing you can do." The context here being that pre-pandemic the benefit of an LP investing in a private equity fund was that achieving low-to-mid teen returns in a senior part of the capital structure wasn't going to happen – thus if you were looking for low-to-mid teen returns, not mid-to-high single digit returns, you'd need to go down the stack to equity (more risk, more return, etc.). Whereas now some PE funds are struggling to even realize low teen returns while those holding senior parts of their PortCo cap stacks are sitting fat and happy getting low teen returns due to the higher rates environment for, obviously, assuming much less risk than those holding equity. [Matt Levine has a fun little piece](#) on this that's worth reading if you have time.

Anyway, here's another way to think about this: with the Term Loan you're getting the same level of priority as with the SSNs, the same level of rate if SOFR is at or under 1%, *and* an effective (not literal) call option on the future of rates. In other words, if rates rise, and the credit spread remains the same, then the SSNs will be clipping the same amount of coupons and the SSNs will fall in price. However, if rates rise the TL will earn more interest *and* the price will be insulated from the rates move because the level of interest earned by holders ebbs and flows with the rates tide.

So, if the Term Loan and the SSNs are hypothetically trading at the same price when we're looking at them (remember: the question assumes we're looking at the Term Loan and the SSNs when SOFR is 50bps, not over 500bps as it is now) then this effective call option on rates rising would be free of charge (read: you're getting the potential for upside without having to pay for it here).



And the rationale for it being free of charge would be that market participants don't see rates rising significantly, certainly not over the floor level, before the maturity of the Term Loan so the option has no real value. This may seem implausible but look at Fed Funds projections from 2020 – the consensus was *very much* that rates would stay well below 1% for years to come and there were situations similar to the setup in this question in the depths of the pandemic due to the belief that we wouldn't see a rates environment where SOFR inched above the floor in the near-term.

Note: Another little caveat I'll add to this answer is that if we *aren't* wedded to the idea of holding either the Term Loan or the SSNs until maturity then it'd make even more sense to invest in the Term Loan in such a low rates (SOFR = 50bps) environment. Because if rates *do* rise from their rock-bottom level in the future the price of the SSNs will fall (maybe a lot!) as the yield adjusts but the TL price will, all else equal, stay relatively stable as its coupon will ebb and flow with the rates tide. For example, imagine if you had bought Coca Cola's 2.5% Notes due in '51 back in '20. There was a time, in late-2020, when they traded *above* par, with a yield below 2.4%, but now they trade in the mid-60s with a 4.6% yield. Needless to say, this isn't because Coca Cola became distressed – their credit risk isn't much different than when they issued these Notes and the yield is still absurdly low. But with the rapid rise in rates – resulting in the 30-year treasury that has *no* credit risk now being over 4% – the yield on the Notes had to adjust. If we were holding these Notes until they mature, then none of this matters too much – we can keep clipping our coupons and we'll get back par in 2051. But if we had to sell then now, well, we're not going to like the results.

1 Let's imagine there are two companies (ABC Inc. and XYZ Inc.) that both have \$100mm in EBITDA and that both have \$500mm in Senior Secured Notes. What are some reasons these SSNs could be trading at different levels?

This question is purposefully vague because the interviewer wants you to just rattle off some potential reasons (there are a lot of them!), so let's do that...

First, the obvious initial thought should be that there could be a simple yield explanation here. For example, the SSNs of ABC could have been issued in a lower rates environment with a lower coupon, and the SSNs of XYZ could have been issued in the now higher rates environment with a higher coupon. So, if both companies are almost identical outside of this then the SSNs should have more or less equal yields, reflecting more or less equal credit risk, but in order for that yield equalization to occur the price of ABC's SSNs will per se need to be lower (price down, yield up).

Second, we haven't been told if these SSNs represent the entirety of each company's capital structure – and that's kind of important! For example, if both ABC and XYZ have an EV of \$550mm but ABC has additional debt residing pari or even senior to their SSNs then you'd expect ABC's SSNs to be trading lower than XYZ's reflecting its shakier (perhaps *much* shakier) recovery prospects. In fact, even if the additional debt in ABC's capital structure were all junior, so the SSNs were still first in priority, the interest expense of this additional debt will weigh on the actual FCF of the company and be a modest (or more than modest) drag on pricing as it'd make the financials of ABC worse and thus the future prospects of ABC shakier (read: more uncertain / variable).

Third, we've been told nothing here about the actual financials of either company beyond the fact that they have the same level of EBITDA. But that's a bit like being shown only a car's engine and then having to guess if the car is worth \$70,000 or \$140,000 – it could be either! The easiest way to quickly rattle off some financials that could sway pricing is to think about this from a liquidity roll-forward perspective. Because we know that if one company has a much healthier (read: sustainable) looking runway then, all else equal, the debt should be trading a bit (or a lot) better.



So, we know that EBITDA is the same for these two companies but think about all the things that could change the FCF picture here. Maybe one company has lots of addbacks, so we'll start at a higher Adj. EBITDA; maybe one company has a lower effective tax rate for some reason; maybe one company has lower capex; maybe one company has lots of liquidity so has more time to turn things around even if the FCF of both companies happens to be negative; etc.

Fourth, we haven't been told anything about the trajectory of the companies. Instead, we've been told that, as of this exact moment, their EBITDA is the same. If both SSNs are trading below par but ABC has contracting revenue and margins then, all else equal, its debt will probably be trading lower than XYZ's based on anticipated future deterioration (likewise if one company has lots of contingent liabilities). Conversely, maybe both SSNs are trading below par but XYZ seems to be turning the corner and there's a belief that they're better situated to "grow into their capital structure" so the market is pricing the debt a bit better on that assumption.

Note: For the above three points, the "shaker financials" mentioned could be a modest (or more than modest) drag on pricing even if the SSNs still look well-covered. You see this frequently where well-covered secured debt, that everyone agrees should get a full recovery if filing occurs in the future, trades down (maybe not too much as it is still well-covered now) because a company with shakier financials simply injects a bit of uncertainty into the mix and, all else equal, investors will want a bit more yield (return) for sticking with a company with a more *uncertain* future (as that more uncertain future raises the probability, even if still remote, of not getting full recovery).

Fifth, we have no insight into the two corporate structures here (e.g., which entities the SSNs were issued out of and where value resides) or what makes these SSNs secured to begin with (we can assume they're secured, but we have no idea by what). As in At Home, two secured notes can be quite different based on where in the corporate structure they were issued and what backs them.

Sixth, we would need to take a look at the debt docs (indentures) here. Maybe both of these companies have no additional debt, but one has loose docs that allow for significant pari or structurally senior debt to be raised or for the company to shift value away from holders through an unsub transfer (in both cases leading to a more insecure position for holders). Likewise, maybe when we look at the docs we see that one SSN benefits from a better call schedule (see pp. 16).

Seventh, we haven't been told here where comps are trading at – or even if these two companies have the same comps (maybe they're in totally different industries!). Needless to say, if ABC has a 9x multiple and there's no other debt then, sure, if the coupon is low it may be trading below par due to our current rates environment but there's little doubt that the SSNs will get back par unless ABC falls apart. Whereas if XYZ has a 5x multiple, so we're right at the recovery borderline, and FCF is negative then we'd expect their SSNs to trade lower, all else equal, based on the more uncertain recovery prospects if XYZ continues deteriorating and files in the future.

Note: It's important to keep in mind that in the real world not all of the above will i) inform pricing or ii) work in the same direction. In the end, all pricing is a kind of combination of factors and there are a million caveats here but this is my best attempt at giving you a laundry list of things to reel off. Today, much of the weird pricing in investment grade (IG) markets (e.g., some Coca Cola Notes issued in '20 now trading in the 60s!) is simply due to yield dynamics (read: low coupon fixed-rate debt issued in a low rates environment per se having to fall in price now that we're in a higher rates environment). However, as a general rule, as companies begin to encircle distress then things that will inform the company's probability of filing and the subsequent recovery value will begin to move pricing more as opposed to the machinations of the broader rates environment.



- 1 Let's say there's a company with a \$500mm Term Loan at S + 7% (assume SOFR is at 1%) that has Adj. EBITDA of \$50mm, opex of \$30mm, capex of \$25mm, cash taxes of \$10mm, a working capital increase of \$5mm, and cash on the balance sheet of \$60mm. Given all this, what's the annual cash burn and runway here?

If you've already read the Case Study Guide then this should all look somewhat familiar as it's more or less the same situation I walked through in the second step of the case but broken out here into a standalone question. So, the purpose of this question is just to get you to do a liquidity roll-forward without, like, actually invoking the phrase. (EVR loves asking this kind of question.)

Therefore, to do a little liquidity roll-forward we just need to start with Adj. EBITDA, work our way down to FCF, and add back the amount of cash on the balance sheet – then, as the name implies, roll this forward until we run out of cash and hit the end of the proverbial runway.

To this end, the only thing we need to figure out before getting started is our annual cash interest which – assuming SOFR stays static through our projection period – will be \$500mm * 0.08 or \$40mm. So, with that found, if we assume everything remains static outside of the cash balance through the projection period then we'll be at the end of the runway in the third year (more specifically, around 40% of the way through the third year since we start with \$10mm of cash and have an annual cash burn of \$25mm).

Illustrative Liquidity Roll-Forward

	2024E	2025E	2026E
Adj. EBITDA	\$55.0	\$55.0	\$55.0
Less: Capex	(25.0)	(25.0)	(25.0)
Less: Working Capital	(5.0)	(5.0)	(5.0)
Less: Mandatory Amortization	-	-	-
Less: Cash Taxes	(10.0)	(10.0)	(10.0)
Less: Cash Interest Expense	(40.0)	(40.0)	(40.0)
Free Cash Flow	(25.0)	(25.0)	(25.0)
Plus: Excess Cash Available	60.0	35.0	10.0
Ending Excess Cash Available	\$35.0	\$10.0	(15.0)

- 1 What are some reasons this runway may not *really* be accurate – or, put another way, a bit too simplified?

I mean, there are a lot of reasons. However, here are some of the big ones...

First, let's start at the top: we're assuming here that our Adj. EBITDA is constant throughout the projection period (pretty remarkable for a company that otherwise is burning quite a bit of cash!). Chances are, if we had the historical financials, we'd see that Adj. EBITDA has been on the decline for a while – so in our base case it would probably make sense to mirror the Adj. EBITDA trend (e.g., 2% YoY declines or whatever as opposed to it being static). Also, it'd be worth taking a few minutes to understand what exactly we're adding back to arrive at Adj. EBITDA to begin with.

Second, when FCF is negative and the end of the runway is quickly approaching then a company, naturally, will begin to cut costs where it can. And for companies with quite a bit of capex all eyes begin focusing there. So, unless the capex figure above is maintenance capex (e.g., the bare minimum needed to not begin destroying value through having machinery blow up from lack of repair, etc.) then there's probably some amount of capex that can be, and would be, trimmed.



Third, the question says there's a \$500mm Term Loan at $S + 700\text{bps}$ and that SOFR is 100bps . Fine. However, the nature of floating rate debt is that it, uh, floats. So, it could and probably *would* move up or down over the three-year projection period. And if it did the level of cash interest expense per year would, obviously, fluctuate and change around how much runway we have (unless the company hedged out their interest rate exposure – many didn't through this last cycle because they operated under the assumption that we'd be in a low rates world for years to come).

Fourth, the Term Loan may amortize over time (e.g., 1% per year) and we aren't reflecting that here – but, if we were to, it'd also lower our FCF as that's cash out the door.

Fifth, the one constant of working capital is that it fluctuates – and here we're assuming a static figure (one that draws down cash, no less) over the entire period. Chances are, as the company begins circling the drain, every string would be pulled to try to limit working capital increases.

Note: There are some other obvious ones like if the amount of tax actually makes sense here and would be static throughout, if cash on the balance sheet is earning any interest, etc. But the five above are the more glaring simplifications.

- 1 Imagine that the company we discussed above doesn't have \$60mm of cash on their balance sheet. Instead, they have \$10mm of cash on their balance sheet plus a \$50mm revolver with \$0mm drawn (so their liquidity appears to still be \$60mm). How, if at all, would this change their runway?**

The short answer is that it would shorten their runway (read: the company would file sooner all else equal), the only question is how much it'd shorten it by. This is for two main reasons...

First, in the last example the company was filling their negative FCF hole with the cash on their balance sheet – never ideal, but it's just cash. However, in this case their FCF hole will be filled through drawing the revolver more and more – and, obviously, this is a form of debt that'll come attached with some interest rate (e.g., $S + 375\text{bps}$ or whatever). Therefore, as the company draws down their revolver more their cash interest expense will begin inching up and shorten their runway yet further. It's a (bad) self-reinforcing cycle. (The company could fill a bit of the FCF hole with the cash on its balance sheet but as we've discussed before the company needs to have a bit of cash floating around to, like, operate so there wouldn't be much that could be used).

Second, the revolver could have a test that's been tripped – or could soon be tripped – thereby capping the amount of the revolver that can be drawn (even though it *looks* like it's fully available right now). For example, in the Double Dip Guide we talked about Trinseo. In their \$375mm revolver that was *undrawn* pre-transaction there was a 3.5x first-lien net leverage ratio test that basically said if the revolver was at least 30% drawn on the last day of a quarter then the test would be triggered (this is called a "blocker" and the details will be found in the credit agreement).

Since Trinseo's first-lien net leverage ratio was already (much) higher than 3.5x this meant that it could only draw on its revolver up until the point before that test would be triggered (30%). Therefore, the *actual* amount available under the revolver was $\$375\text{mm} \times 30\% = \112.5mm (well, *technically* it was \$100.5mm because you needed to net off the letters of credit outstanding but that's not the important point). Alternatively, if Trinseo had already drawn XX% of the revolver but had, for example, 3.0x first-lien net leverage then they would have been able to continue drawing the revolver only until the point of hitting the 3.5x first-lien net leverage ratio – and this could be an amount of revolver availability well below the amount that should theoretically be available.



So, in our little example above, there's theoretically \$50mm available but because of a blocker in the credit agreement the future draws may be capped at some (much) lower amount to maintain compliance. Therefore, it *looks* like we have \$60mm of total liquidity available but, depending on the docs, there might only be \$30mm via \$10mm of cash and \$20mm of revolver availability.

As I've written before, all credit agreement are unique snowflakes and this is just one example of a test that can cap revolver availability below what, at first blush, it appears that there should be. (Also, as discussed on pp.19 in the Bonus 2022 Guide, for an ABL its availability could be reduced simply due to the borrowing base declining). Anyway, don't worry about the specifics too much – rather, keep in mind the idea that revolver availability is *usually* straight-forward to calculate but in a distressed situation the amount of availability may be a bit, or a lot, less than it appears.

1 Imagine that a company has \$625mm of Senior Notes outstanding. It then proposes a transaction whereby holders can swap their Senior Notes into 2L Notes at a 50% exchange rate and into equity at a 50% exchange rate (there are currently 100mm shares outstanding and the share price is \$5). Assuming that the participation rate is 80%, what would the post-transaction equity splits be?

So, here we have a pretty simple debt swap with an equitization component. If you were among the 80% of Senior Noteholders who participated and happened to hold \$100mm of the Senior Notes then you'd end up post-transaction with \$50mm worth of the new 2L Notes and \$50mm worth of equity.

To figure out the equity splits (e.g., how much of the equity will be owned by the exchanging noteholders and how much will be owned by the pre-existing shareholders post-transaction) we can break it down into a few steps.

First, we need to figure out the face value amount of Senior Notes being exchanged, the amount of exchange consideration in equity that they're receiving, and how many shares will need to be issued to effectuate this.

Since we've been told that the participation rate is 80% that means that $\$625\text{mm} \times 80\% = \500mm of Senior Notes are participating in the exchange. And since we've been told that the exchange consideration comprising equity will be 50% of the face value exchanged, this means that we need to provide equity value that's equivalent, at time zero, to $\$500\text{mm} \times 50\% = \250mm to exchanging noteholders. Therefore, assuming that the new shares are issued at the current trading price, this would be $\$250\text{mm} / \$5 = 50\text{mm}$ shares.

Second, if we think about the actual equity splits then we're moving from having 100mm shares outstanding pre-transaction to 150mm post-transaction. So, 33.33% ($50\text{mm} / 150\text{mm}$) of the shares will be held by exchanging noteholders and 66.67% ($100\text{mm} / 150\text{mm}$) will be held by the pre-existing shareholders.

Here's one way to think about it: pre-transaction we said there were 100mm shares outstanding trading at \$5 thereby giving us an equity value of \$500mm. Now, we know this transaction is significantly delevering because we're only swapping 50% of the face value amount of participating SNs into new debt (the 2L Notes) and are equitizing the remaining 50%. Therefore, if we assume that EV remains constant the \$250mm in debt reduction from the equitization must be offset by a \$250mm rise in equity value (in other words, EV doesn't change the day after the transaction relative to the day prior to the transaction – but the *composition* of EV does).



So, this transaction wouldn't, in theory, change the current share price as we now have 150mm shares outstanding and if we multiply this new share count by the pre-transaction share price of \$5 that brings us to \$750mm which is equal to our post-transaction equity value referenced above (although, of course, this is all theoretical: there could be a bump in the share price due to increased belief in the company being able to turn things around with a now delevered balance sheet, or there could be a fall as folks worry about the company opening the door to dilution happening down the road in future transactions, etc.) Anyway, those nuances aside we should expect that the 50mm shares post-transaction belonging to exchanging Noteholders, multiplied by the current share price, will give us \$250mm (as it does) and that the 100mm shares post-transaction belonging to the pre-existing shareholders, multiplied by the current share price, will give us \$500mm (as it also does).

Exchange Year	2024E
Shares Outstanding	100.0
Current Share Price	5.0

(\$ in millions)

Equity Splits	
Total Senior Notes Exchanged	500.0
Exchange Consideration % in Equity	50.0%
Consideration Value in Equity	250.0
Shares Issued at Current Share Price ¹	50.0
Dilution at Close	33.3%

Share Count	Pre-Reorg	Adj.	Pro-Forma
Exchanging Noteholders	-	50.0	50.0
Existing Shareholders	100.0	-	100.0
Total	100.0		150.0

Share %	Pre-Reorg	Adj.	Pro-Forma
Exchanging Noteholders	0.0%	33.3%	33.3%
Existing Shareholders	100.0%	(33.3%)	66.7%
Total	100.0%		100.0%

- 1 Imagine the following transaction: a new \$200mm 1L TL is raised and the pre-existing \$500mm Senior Notes are exchanged into new 2L Notes at a 90% exchange rate (assume an 80% participation rate). Assuming there are \$10mm in transaction fees and that the 1L TL proceeds are used, in part, to repay \$100mm drawn on the pre-existing revolver, what would the sources and uses look like here?

Here we have a pretty simple transaction structure: we're raising \$200mm of new 1L debt to, in part, paydown a revolver by \$100mm and we're also exchanging some Senior Notes into new 2L Notes (the exchange is being done at a discount but those participating will prime the non-participating SNs). Since 80% of Senior Noteholders are participating, the total amount of Senior Notes being exchanged will be \$500mm * 80% = \$400mm. And, since the exchange rate is 90%, those participating will receive \$360mm in new 2L Notes.

So, most of the sources and uses here should be pretty intuitive (e.g., the paydown of the revolver and the transaction fees are obviously a use, the new 1L TL raised is obviously a source, etc.). But the reason I cooked up this little question is to illustrate that in this transaction the exchange is being done – as most exchanges in distressed situations will be! – at a discount, so there's some discount capture here (in other words, the amount of consideration an exchanging holder is receiving is less than the face value of the debt they're exchanging). This discount capture will be on the sources side and will be the face value of the SNs participating (\$400mm) minus the new consideration given to them (\$360mm) or \$40mm.

Sources	
New 2L Notes	360.0
New 1L TL	200.0
Discount Capture	40.0
Total Sources	600.0

Uses	
Exchange Financing Fees	10.0
Revolver Paydown	100.0
Cash to Balance Sheet	90.0
Exchanging Senior Notes	400.0
Total Uses	600.0



- 1 **Imagine that there's a company with two segments: nuts and bolts. The nuts segment has \$50mm of EBITDA and comps are valued at a 6x multiple. The bolts segment also has \$50mm of EBITDA but comps are valued at 8x. The entire business has \$20mm of opex that's not captured within the two segments. Given this, what's the TEV?**

Let's take a breather from restructuring questions for a minute. Here we're just going to add together the EBITDA of each segment to find the total EBITDA: \$100mm. Now, since we've been told the entire business has opex of \$20mm the whole business has EBITDA of \$80mm. (We're assuming here that there's no EBITDA contribution outside of the two segments, there's just some random opex from generally running the business that's not being captured within the segments).

Since the EBITDA contribution of each business is 50% of the whole business, then with the limited info we have it makes the most sense to split the whole business EBITDA equally (\$40mm for nuts, \$40mm for bolts). So, if we then apply our multiple of 6x and 8x, respectively, we'll get a TEV of \$240mm + \$320mm = \$560mm.

- 1 **Imagine that there's a company with a healthcare division that was sold for \$100mm at a 5x multiple. Is this a deleveraging transaction for the company? Feel free to ask for more details as needed.**

First, the EBITDA of the healthcare division was, obviously, \$20mm. But to figure out if this transaction was deleveraging we need to know the company's overall EBITDA and level of debt.

If we were told that it was \$100mm and \$400mm, respectively, before the transaction and we assume the proceeds from the sale went to paying down debt then we went from the company being 4x levered (\$400mm / \$100mm) to being 3.75x levered (\$300mm / \$80mm). Therefore, the transaction reduced overall leverage by 0.25 turns, so was modestly deleveraging.

- 1 **There's a lot of talk these days about Liability Management Exercises (LMEs). So, what's this talk all about? What are these?**

There are many bad habits that lawyers have – but one of them that has knock-on ramifications on all of us is a pension for coining acronyms that serve no purpose and then bludgeoning everyone into using them. (Literally [trying to make fetch happen](#) – perhaps this is one of the inevitable and disastrous consequences of millennials now getting older and getting into more senior positions?)

LME, short for liability management exercise, is one of these acronyms. It's hard to believe there's a turn of phrase, and an acronym derived from it, that is both so boring *and* so devoid of explanatory power. At least when Congress creates absurd acronyms out of their legislation they have a bit of fun with it. The Sequestration Tied to Member Pay (STOMP) Act. The Servicemember Assistance for Lawful Understanding, Treatment, and Education (SALUTE) Act. The Working to Encourage Community Action and Responsibility in Education (WE CARE) Act.

I've tried my best to avoid invoking this acronym for a few years, hoping against hope that it would retreat back into the ether a bit because it *is* more or less useless. But it's now become lodged into the lexicon so, for the sake of completeness, let's talk about it as best we can.



First of all, there's no definitive definition of LME, nor is there a uniform usage of LME. It's kind of one of those things that's thrown around in some circles, without fear or favor to the context, and that everyone nods along with because they have *some* vague notion as to what it's referring to.

Most use LME as a catchall to refer to the more creative (read: aggressive) types of out-of-court transactions that I've discussed at length over the past few years (e.g., drop-downs, non-pro rata uptiers, and double-dips).

For example, if you were to use LME in a sentence it'd be something like, "There were LMEs done by At Home, Trinseo, and Frontier in 2023 that'll hopefully work out better than the LMEs done by Envision and Incora in 2022 since both of them ended up filing in 2023."

When used this way to refer to a pretty narrow set of transaction types it's not an entirely useless term. It's more efficient than saying drop-downs, non-pro rata uptiers, etc. over and over again when talking in broad terms about these *types* of transactions. For example, it's a lot easier to say, "LMEs require a careful analysis of the debt docs, although interpretations may vary and in most LMEs litigation from those 'harmed' should be expected no matter what – it's par for the course and just the cost of doing one of these transactions."

However, some, and this really grinds my gears, have adopted LME to describe any type of transaction that allows a stressed or distressed company to obtain liquidity, capture discount, and/or extend maturities. So, this would even make a vanilla amend-and-extend an LME. This is a ridiculous situation since, like, we have the term out-of-court transaction that's been used for decades as a catchall to refer to all the types of transactions that can, like, happen out-of-court.

In the end, it doesn't matter. This is just a catchall term that is most often used almost as a euphemism for types of transactions that are going to be litigated. However, I'm not a lone voice in the wilderness vis-à-vis my soft-hatred of the acronym. Because now you'll hear people say stuff like, "We think an LME is possible here and is worth exploring" and everyone is like, "Uh, sure, right, of course, *fantastic*, but what do you *mean*? What kind of transaction are we specifically talking about here?"

Anyway, in an interview if anyone uses the acronym then they'll almost certainly have the narrower definition in mind (e.g., a catchall for drop-downs, uptiers, double-dips, and their ilk) and if you're asked what it means (doubtful) then use the narrower definition.

But rest assured that if you've read through the Serta Postmortem Guide and the Double-Dip Guide then you already know what the proper *types* of LMEs are (even if you were unfamiliar with the acronym until now) and that's what matters. It's a catchall – but, as it happens, one that people can't quite figure out what it's meant to be catching (presumably a prerequisite of a catchall is that people should mostly agree on what it's catching, or at least agree on how to use it in a sentence, but I digress...).

Note: Some use Liability Management Transaction (LMT) instead of LME but the definition, insofar as one exists, is the same.

- 1 Let's imagine there's a company that, at the end of Q2, had LTM EBITDA of \$10mm and was trading at a 5x multiple. Then, at the end of Q3, had LTM EBITDA of \$15mm and was trading at a 6x multiple. Given this, what percent of the change in EV do you think was likely due to changes in EBITDA vs. multiple expansion?**



Before getting too ahead of ourselves, let's first figure out the EV change here. At the end of Q2 EV was \$50mm and at the end of Q3 EV was \$90mm, representing a \$40mm increase. Now, there are a few ways to think about this and there's no "right" answer but I think the best way is to break this delta into three sections: the increase that we *know* is due to the EBITDA increase, the increase that we *know* is due to the multiple expansion, and the increase that is a bit iffier (in other words, can't be attributed in a clear-cut way to either).

The way to think about the EV increase that we *know* is due to the EBITDA increase is that it'll be the increase in EBITDA between the quarters multiplied by the Q2 trading multiple (because if the trading multiple didn't expand between quarters, then we *know* that the entire EV increase would be due purely to the EBITDA expansion). Therefore, in this case, the EV increase would be \$5mm * 5x = \$25mm or 62.5% of the total EV increase of \$40mm.

The way to think about the EV increase that we *know* is due to the multiple expansion is that it'll be the multiple expansion multiplied by EBITDA from Q2 (because if EBITDA didn't expand between quarters, then we *know* that the entire EV increase would be due to the multiple expansion). Therefore, in this case, the EV increase would be \$10mm * (6x - 5x) = \$10mm or 25% of the total EV increase.

The way to think about the EV increase that is iffier is simply the amount we know EV went up by (\$40mm) subtracted by the EV increase that we know came from the EBITDA increase (\$25mm) and the multiple expansion (\$10mm) or \$5mm (in other words, 12.5% of the total EV increase). Alternatively, we can think about it as being the EBITDA expansion that occurred multiplied by the multiple expansion that occurred \$5mm * 1x = \$5mm.

Now, one can make an argument with how to split this EV increase that we're saying is attributable to both the EBITDA expansion and the multiple expansion and it won't matter for interview purposes. Personally, I think it's easiest to split the baby – so, since \$5mm represents 12.5% of the total EV increase we'd just tack on 6.25% to both the EBITDA expansion (62.5% + 6.25% = 68.75%) and the multiple expansion (25% + 6.25% = 31.25%).

However, one could try to make an argument that the multiple expansion is a sign of the market *rewarding* the company for its fast growth or cost discipline or turnaround or whatever caused the EBITDA increase – therefore more of the 12.5% (maybe all!) should be tilted toward the EBITDA expansion bucket. This line of argument would have a bit more weight if comps didn't see as much of an uptick in EBITDA *and* had similar multiples to the prior quarter, so one could then say that this company had outsized EBITDA growth and was rewarded with outsized multiple expansion as a result (in other words, this isn't just a rising-tide-lifts-all-boats scenario).

1 Let's imagine that ParentCo has \$200mm of Senior Secured Notes and holds the equity of SubCo. Meanwhile, SubCo also has \$200mm of Senior Secured Notes that have the same maturity and coupon as ParentCo's SSNs. If you were trying to decide which to invest in, what are some of the things that you'd think about?

The purpose of more vague questions like this one is simply to make sure that you've internalized a few of the core rx concepts we've talked about and know when to draw on them. So, let's briefly walk through some of the major things that should immediately spring to mind...



First, we would want to figure out where the value actually resides in the corporate structure as, all else equal, it's better to reside where lots of value is located (e.g., if there's \$100mm of value at the ParentCo and \$200mm at the SubCo then, all else equal, one would rather be at SubCo).

Second, since this is a classic HoldCo / OpCo structure, we would want to know what kind of guarantees (if any) are at play here (e.g., maybe ParentCo has no or limited value, as mentioned above, but has a secured guarantee from SubCo thereby rendering the ParentCo SSNs pari).

Third, we would want to turn to the debt docs and figure out the level of pari debt that could be incurred at either the SubCo or ParentCo or if there's a difference or if it matters. All else equal we want to avoid the potential of having more debt "compete" for recovery alongside us or above us, but maybe the SSNs reside pari to each other (due to a guarantee) so whatever additional pari debt can be incurred would have the same impact on both SSNs. (This is a more complicated rabbit hole and it doesn't make sense to start talking about all the edge cases here. Rather, it's just worth highlighting that there could be a difference here in how much debt could be incurred alongside you depending on the structure and where you reside within it.)

Fourth, we would want to consider the above all in the context of the actual price of the ParentCo and SubCo SSNs. Because the question is asking us what we'd *invest* in, not what we think will get the highest recovery. For example, let's say we do a deep dive here and determine that the SubCo debt is fully covered – in other words, there's no way it'll not get full recovery if filing occurs. However, we aren't the only ones who think this, and the debt is trading around par with a coupon of 10%. So, this wouldn't be a bad investment – no one sheds a tear over a safe 10% IRR. But maybe we determine that the ParentCo SSNs, that also have a 10% coupon, are liable to get around a sixty-cent on the dollar recovery if filing occurs. However, the price of the ParentCo SSNs are around forty. In this case, we're taking on more risk as the value underlying the SSNs could deteriorate further – but if we're comfortable with our assumptions and think the market is just fundamentally mispricing these SSNs then it'd make more sense to invest in these (operating under the assumption that our fund has the mandate and risk tolerance for an investment that, at a minimum, will be more volatile through the hold period and have a higher variance in outcomes).

So, this is a reasonable starting point. As I've mentioned many times before, two things can have very similar names but have very different trading levels and end up providing very different IRRs – and the reason mostly boils down, at a base level, to some combination of the factors above.

1 Let's imagine that a company has a balance sheet consisting of \$50 of inventory and \$50 of equity. It then sells \$50 of inventory for \$100 with 50% of sales being cash and 50% being on credit. Can you walk me through the three statements?

So, beginning on the income statement we have revenue of \$100 and COGS of \$50 leaving us with EBT of \$50. Assuming a 20% tax rate, we'll have net income of \$40. Turning to the cash flow statement, we begin up \$40. Now, inventory has declined by \$50 (source of cash) so we'll add \$50 but accounts receivable has increased (use of cash) by \$50, so we'll subtract \$50 – since these offset our cash flow from operations is up \$40 and there are no other changes on the cash flow statement. Finally, on our balance sheet we'll have cash up \$40, inventory down \$50, and accounts receivable up \$50. So, our asset side of the balance sheet has increased \$40 and is \$90 overall (\$40 cash, \$0 inventory, \$50 accounts receivable). On the liabilities and shareholders' equity side the only change is \$40 of net income flowing into retained earnings (therefore, the liabilities and shareholders' equity side stands at \$90 from the \$50 in pre-existing equity and the increase in retained earnings of \$40).



- 2 Let's imagine that a company – with a capital structure consisting of a 1L Term Loan and 2L Term Loan – does an uptier that involves participating 1L lenders exchanging their holdings into a new super-priority second-out tranche and funding a new-money super-priority first-out tranche. Therefore, after the transaction there are now three (instead of one!) tranches ahead of the 2L TL: the first-out new-money tranche, the second-out tranche, and the pre-existing 1L TL tranche if we assume that not all pre-existing 1L lenders participated in the transaction. Given all of this, is there any plausible scenario where the price of the 2L TL, that's now been pushed back in priority so much, could actually be higher immediately after the transaction compared to where it was immediately before the transaction?

We've talked a lot in the past about non-pro rata uptiers and the common thread between them has been that non-participating lenders and those even further back in the capital structure have had the value of their holdings eviscerated by these transactions – something that shouldn't be too surprising since we're talking about debt that was, for example, first-lien or second-lien effectively becoming third-lien or fourth-lien or even fifth-lien behind millions in fresh debt which, uh, is probably not going to be great for their recovery if the company files down the road.

This is (obviously!) why these transactions get litigated so much. If these transactions most often resulted in the trading price of the debt held by those that *didn't* participate increasing post-transaction then no one would be complaining too much. However, one can imagine a scenario where the 2L debt in this kind of scenario *could* trade higher even though that seems a bit counterintuitive. In fact, we've seen this occur (albeit it's *far* more common for those not participating in an uptier to take a bath than for their holdings to somehow increase in value).

So, let's talk briefly about Pretium Packaging (sponsor backed: Clearlake). They had a \$1,300mm 1L TL due Oct '28 (L + 400) and a \$350mm 2L (L + 675) due Sept '29. In their transaction things were a bit different than in Serta, Boardriders, Incora, etc. because *all* 1L holders were given an opportunity to exchange their existing 1L debt into a new super priority, *second-out* tranche at an exchange rate of 93.6% (this new second-out debt is S + 460 and due Oct '28).

However, in order to participate in the exchange the existing 1L holders would also need to participate, pro-rata, in funding a new \$325mm super-priority, *first-out* tranche due in Oct '28. So, 1L holders that *didn't* participate would be primed by the first-out new-money tranche and the second-out tranche comprised of those that did participate, and 2L holders would be primed by the first-out tranche, second-out tranche, and those 1L holders that didn't participate in the transaction (in the end, the transaction was backstopped pre-announcement by around 85% of 1L holders and almost everyone participated).

Note: So, to be clear, the transaction was a *pro-rata* uptier vis-à-vis the existing 1L TL as everyone was allowed to participate if they wanted to. However, from a 2L holder's perspective, which is what we care about here, the impact was no different than in a classic non-pro rata uptier (being primed by multiple new tranches of debt without the ability to participate in the transaction).

Anyway, over the summer (before the transaction) the 2L TL traded down and down and then down some more – from the high 50s to the high 20s in the days before the transaction. This was because the company was burning around \$10-20mm of cash per quarter, and their liquidity was drying up (\$10mm of cash on the balance sheet, probably around \$20mm available on the ABL).



So, the reason the 2L traded ever downwards was that everyone began to recognize the 2L was going to receive little recovery in the eventual filing, and that filing was preordained with how little runway was remaining (it was no longer a matter of *if* filing would occur, merely a matter of *when*).

But post-transaction the 2L TL traded up from the high 20s to the high 30s. Now, one could argue this rise is partly reflecting the fact that there's more optionality now. The company has more breathing room, maybe it'll turn things around! However, there are few (to be charitable) that *truly* believe in the future of Clearlake's distressed PortCos (including Wheel Pros that I discussed in the Double-Dip Guide). Everyone knows the endgame is probably still filing – just not this year.

I mean, sure, the 2L probably traded up some because of hopes for a better future. But the 2L TL trading up post-transaction was mostly because the transaction brought in a few hundred million in fresh cash, and this cash can be used to fill the negative FCF hole for a lot of quarters. So, because there are no near-term maturities, Pretium has bought itself quite a bit of time and this means that those L + 675bps interest payments will keep flowing to holders for much longer than the market was anticipating pre-transaction (since, again, they had more or less run out of cash).

In other words, pre-transaction the 2L TL was trading down to such low levels because holders thought they would get a very low recovery (probably a smidge of post-reorg equity) upon filing *and* that filing would occur in the very near future (so there wasn't going to be much more interest coming their way pre-filing). Now, post-transaction, there's no doubt that the 2L TL is in a *worse* position from a recovery-perspective (they've just been primed by hundreds of million in new debt and no one has much faith in the company magically turning things around!) *but* the company will be able to pay a lot more interest to the 2L TL than before. (If you're buying a L + 675bps 2L in the 20s then it doesn't take too much cash interest to earn a nice return even if the recovery upon filing is even lower than before – just so long as that filing is sufficiently far into the future.)

Now, if the 2L TL had been trading in the 80s pre-transaction then that would've been reflecting the fact that even though the company was likely to file soon given their limited runway – and thus there wouldn't be much more, if any, cash interest earned by holders – the recovery value would've been anticipated to be high (e.g., in-line with the trading level). So, all else equal, upon this kind of transaction being announced you'd expect the 2L TL to *fall* in price here because all of a sudden, sure, the company will be able to stay afloat and keep making interest payments for longer *but* the fact that the 2L TL has been heavily primed by new debt will result in their recovery value taking a big hit unless the company turns things around soon and EV increases significantly.

In other words, the benefit of the continued interest and enhanced optionality will be more than offset by the fall in the anticipated recovery value, whereas in the actual Pretium situation the continued ability of the company to make interest payments to the 2L, along with the increased optionality the company has from a now much more elongated runway, more than offsets the fall in the anticipated recovery value for the 2L (since the latter was very low to begin with, thus why the debt was trading down to such low levels pre-transaction when filing seemed imminent!).

Note: Another (more mechanical) situation where the 2L TL could trade up post-transaction would be if the exchange of, in this example, the 1L TL into the second-out tranche was done at a large enough discount that the discount capture created was greater than the amount of new-money funded in the transaction. In other words, if the new first-out debt, the new second-out debt, and the remaining 1L debt that didn't participate in the transaction is *less* than the original 1L TL then that would mean there's obviously less debt ahead of the 2L TL. So, all else equal, that would be a net positive for the 2L TL recovery *and* the new-money would be used to continue making



interest payments to the 2L holders. So, it'd be a win-win from the 2L perspective. (Needless to say, in the real world there isn't that much, or sometimes any, discount capture created in these transactions so those that didn't participate will have lots more debt in front of them than before).

- 2 **Imagine that a company has a 50% chance of their EV being \$1,000mm and a 50% chance of their EV being \$0mm in the near future. The company's total debt consists of a lonely \$500mm Term Loan at $S + 5.0\%$ (assume SOFR is 5.0%). Given this, what should the Term Loan trade at?**

It's tempting here, especially with the acronym "EV" being thrown around, to immediately think of taking the EV (expected value) of EV (enterprise value) or $(\$1,000\text{mm} * 50\%) + (\$0 * 50\%) = \$500\text{mm}$ and then conclude that the \$500mm Term Loan should trade around par right now.

However, let's think about this a little bit more. We're being asked here where the Term Loan will trade and we know that the EV matters from a waterfall perspective. But we also know that above some threshold EV kind of stops mattering. For example, if the company's EV is \$1,500mm or \$1,000mm or \$900mm or \$800mm or \$700mm will that make a difference in where the Term Loan should trade? In all the above the Term Loan should be well covered and thus we'd expect the Term Loan to trade more or less around par.

So, the right way to think about this is not in terms of the expected value of *enterprise value* under these two scenarios but in terms of the expected value of the *trading level* of the Term Loan under these two scenarios. So, in the first scenario, where EV will be \$1,000mm, the Term Loan should trade around par and in the second scenario, where EV will be \$0mm, the Term Loan should trade around zero. Therefore, the expected value of the trading level should be around fifty cents on the dollar because there's a 50% shot that the Term Loan will be worth par in some future scenario and a 50% shot that it'll be worth nothing in some future scenario.

Now, this question is purposefully vague (e.g., we don't know when this binary EV outcome will occur beyond the fact it'll happen in the near future, etc.) so there's only so much we can say here and there's certainly no singular right price. However, as a little head check, think about the basic return structure for a holder of the Term Loan if they had bought \$100mm of the TL for what we're positing the TL is worth: fifty cents on the dollar.

Given the obvious risks here, a holder will want to be compensated for the risk of, like, the Term Loan becoming worthless at some undefined time in the future. Now, if that undefined time is a year from now, and under both scenarios we think holders will earn \$10mm in cash interest before the EV magically shifts in either direction, then under the \$1,000mm EV scenario the investor will earn \$10mm in interest and \$50mm from price appreciation as the Term Loan will trade from fifty cents on the dollar to around par when this magical shift in EV happens. However, under the \$0mm EV scenario the investor will earn \$10mm in interest and lose \$50mm on the Term Loan itself if we're assuming there's no recovery when this magical shift in EV happens.

Therefore, if we assign equal probabilities to both occurring – a 50% chance that we make \$60mm and a 50% chance that we lose \$40mm – then the expected value of our return over this year is \$10mm or 20% ($\$10\text{mm} / \50mm) assuming that we bought \$100mm of the Term Loan at time zero for fifty cents on the dollar. In the current environment where, as mentioned vis-à-vis the Schwarzman quote earlier, one can hold debt that has much less *variation* in the return structure and that nets out low-to-mid teen returns, maybe this isn't too enticing of a return – but it's still not too bad if one has the risk tolerance.



Note: This question is somewhat contrived as it assumes there's full certainty about two wildly different EV scenarios occurring in the near future, so it's important not to read into it too deeply and twist yourself into knots thinking about it in increasingly abstract levels. I'm really just trying to build up a bit of intuition with the return talk above because, in this case, the actual current trading price will largely be informed by the level of return market participants are willing to accept since the outcomes (\$1,000mm, \$0mm) are predestined (and this return needs to "compete" with the kind of returns that can be earned by holding a piece of debt where there's little chance of *not* getting back par and that pays you a pretty healthy amount of interest in our current rates environment). For example, if the TL was trading at 60 – based on the assumptions above about one of the EV scenarios magically materializing in exactly one year – then the return structure would make no sense as the pricing would basically be saying there's a 50% shot of you making \$50mm and a 50% shot of you losing \$50mm. But you want to be compensated for assuming risk, and despite the amount of risk inherent to this hypothetical there's no compensation for risk happening here – you're just flipping a coin and hoping to get lucky.

1 If there's a bond trading below par, rank the following from lowest to highest: coupon rate, current yield, yield to call, yield to maturity, and yield to worst.

So, we know that the coupon rate is $(\text{Coupon} / \text{Par})$ and the current yield is $(\text{Coupon} / \text{Current Price})$ – therefore, we know that the coupon rate must be lower than the current yield since the current price is below par (100) and that both the coupon rate and the current yield must be lower than the YTM, YTC, and YTW since these latter three also take into account the price appreciation that we're getting through holding a bond that's trading below par. (Think back to the "intuitive" YTM method in the Bond Math Guide where YTM is the current yield component *plus* the price appreciation component – since the coupon rate and current yield per se ignore the price appreciation component, they must be lower!)

Now, if a bond is trading below par then the YTC should be higher than the YTM as the YTC will per se involve a shorter duration and, since the bond is trading below par, that means you're (theoretically) getting the price appreciation quicker (no different than how the YTM of a bond trading below par, if one keeps everything else constant, will increase as one makes the number of years to maturity less). Plus, the call price will be some number a bit above par (e.g., 102%) for some period after issuance to try to assuage investors that if the bond is redeemed (read: fully taken out) before maturity it'll be done at a higher price than par. So, there'll be a bit more price appreciation baked into YTC between now and the call date than with YTM where the price appreciation is from the current price up to par between now and the *actual* maturity date.

Note: YTC is normally calculated based on the earliest call date. However, there'll be a call schedule that'll be like NC2 (can't call for two years) then 102% for some period of time thereafter, then 101% for some period of time thereafter, then 100% (par) until maturity.

Note: Now, if we're saying that YTC is higher than YTM then holders should theoretically welcome being taken out early as it'll boost their return all else equal. However, if a bond is trading above par, as opposed to below, this all flips and the YTC will be lower than the YTM. And, if a bond is trading above par, it's likely because the rates environment has shifted lower and/or the company has become a healthier credit (yields down, prices up). This means the company will be incentivized to refi out their existing debt with lower coupon debt, so the call protection ensures that holders who have made a correct bet on the rates environment shift and/or the company's prospects can enjoy their higher coupons without the company coming in and taking them out



too pre-maturely – or, if they are allowed to take them out, to do so at a premium (for a certain period of time) that rewards pre-existing holders a bit above what they'd otherwise receive for their correct bet.

The YTW will be equal to the lesser of either YTC or YTM as it's the lowest *actual* yield an investor could earn assuming there's no default event. I'm saying the lowest *actual* yield here because the current yield isn't the *actual* yield a holder earns because it ignores the price appreciation component a holder earns by holding a bond below par that gets back par or the call price (again, think back to the intuitive YTM formula discussed in the Bond Math Guide where YTM will be the combination of the price appreciation component *plus* the current yield component).

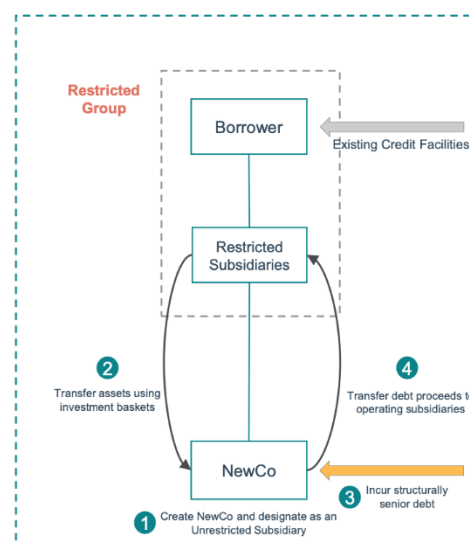
Anyway, in this case the YTM is lower than the YTC thus the YTW will be equal to the YTM. Therefore, in answering the question, you'd say: coupon rate < current yield < yield to maturity = yield to worst < yield to call.

Note: With all of that said above, the estimated YTC formula should be pretty intuitive. It's simply the estimated YTM formula but we're replacing par with the call price and the number of years to maturity with the number of years until the call date: $((\text{Annual Interest}) + ((\text{Price to Call} - \text{Current Price}) / (\text{Years to Call}))) / ((\text{Price to Call} + \text{Current Price}) / 2)$. So, if this is all a bit confusing just take a few minutes to think about the above equation in relation to the estimated YTM formula and it'll all become clear (in the end, the only thing you really need to remember is that you're getting more price appreciation over a shorter period of time).

1 What's a J. Crew blocker and has it become common?

We've talked (a lot) about drop-downs vis-à-vis Serta and also a bit vis-à-vis Trinseo in the Double-Dip Guide. Here's a nice little overview as a refresher...

- 1 Company creates "NewCo" and designates it as an Unrestricted Subsidiary:
 - Because NewCo is an Unrestricted Subsidiary, covenants in the existing credit agreement do not restrict its activities.
 - NewCo can be formed at the time of, or prior to, the transaction.
 - Designation of Unrestricted Subsidiaries may require compliance with certain conditions (such as financial ratio tests and/or default blocker).
- 2 Company then transfers assets to NewCo using available investment and/or disposition baskets in the credit agreement.
- 3 Incur structurally senior debt at NewCo from existing lenders, sponsor, or 3rd party:
 - Quantum of debt raised will be a function of value transferred and market debt.
- 4 Proceeds of new debt up-streamed to the borrower/operating entities to fund cash flow shortfalls and bolster liquidity (and also refreshing investment capacity as returns on investments).



The reality is that preventing a drop-down, similar to preventing a double-dip, is a bit harder to do than preventing a non-pro rata uptier because the crux of the transaction is utilizing the investment capacity that's outlined in the debt docs (as I touched on a bit in a postscript within the Serta Postmortem Guide).



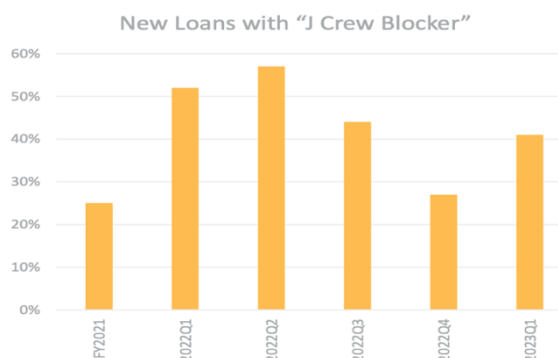
I mean, sure, these could be prevented by taking a more sledge hammer approach and providing less investment capacity in the docs – but the company wouldn't be as interested in this as the investment capacity is critical for other business uses too and would heavily restrict its flexibility (remember that all debt docs are heavily negotiated and a company, all else equal, will opt to go with lenders that are willing to agree to docs that put them in less of a straight-jacket – sometimes even if that results in a slightly higher rate).

Since the “classic” drop-down involved the transferring of intellectual property, and since IP is the crown jewel for many types of companies, the classic J. Crew blocker has been aimed at preventing companies from shifting material IP from the restricted group down to an unrestricted sub (which, if allowed, would then enable the company to incur fresh debt at the unsub and give those new unsub lenders a structurally senior claim on the material IP residing there).

This kind of blocker, to be clear, doesn't prevent drop-downs from occurring writ large – other material assets could still be transferred down. Rather, it prevents one specific type of drop-down that is (was) most common, especially among retailers where the IP is a critical part of the business (in other words, where the brand really *is* the business). So, it's a bit more surgical and something that companies have been more liable to go along with.

There are some different formulations of J. Crew blockers and some are a bit easier to side-skirt than others. But the most traditional basically says, “The Company shall not permit any of its Restricted Subsidiaries to sell, convey, transfer, or otherwise dispose of or license its Material Intellectual Property to any Unrestricted Subsidiary.” (There can also be prohibitions on transferring IP to non-guarantor restricted subs since even though they are bound by the negative covenants of the pre-existing debt docs they don't guarantee the pre-existing debt, so holders of the non-guarantor restricted sub debt would have a structurally senior claim on the IP located there.)

J. Crew blockers have become more common although it's worth mentioning that there's a bit of selection bias here. The healthiest companies (in other words, those that at issuance are least likely to do a drop-down) or those who are least set up to do a drop-down (in other words, those where IP is less central to the business) are those most liable to include it as the company basically has the attitude of, “Well, lenders seem to care about this, so whatever we'll make the concession.”



Note: As mentioned earlier, the term “J. Crew blocker” is a catchall for drop-down protection of any kind – even if that protection is protecting against transactions that are different than the original J. Crew drop-down (e.g., involving material assets of any kind, not just material IP).

Note: In the past we've seen a few J. Crew blocker formulations that prevented restricted subs that held IP from being designated as an unsub but that *didn't* otherwise limit the transfer of IP to an unsub. I touched on designation devilry in the Double Dip Guide, so the loophole in this style of blocker might be immediately obvious to you: the company could simply designate a random (empty) restricted sub as unrestricted and *then* transfer the IP to it after (so, in this case, a restricted sub that held IP would never have technically been designated as unrestricted).



Note: There have been a few credit agreements that have adopted so-called Envision blockers that, like J. Crew blockers, are meant to prevent drop-downs but that take a different approach and were named after a different contentious transaction (Envision) that involved a Frankenstein-esque multi-step drop-down and uptier. These blockers seek to limit the *dollar amount* of assets, not the *type* of assets, that can be transferred to an unrestricted subsidiary, and this is accomplished through only allowing the company to tap the dedicated “investments in unrestricted subsidiaries” basket as opposed to tapping *all* the broader set of investment baskets (e.g., investments in unrestricted subsidiaries basket + general investments basket + leverage-based investments basket, etc.) to effectuate an unsub transfer. Again, see one of the postscripts in the Serta Postmortem Guide where I walked through how Apollo, et al. were limited in how they could amend their proposed drop-down to make it more enticing to Serta due to the limit on the dollar amount of assets that could be transferred to an unsub after tallying up *all* their applicable baskets. (In case it’s not self-evident, this is all so much beyond what you need to know.)

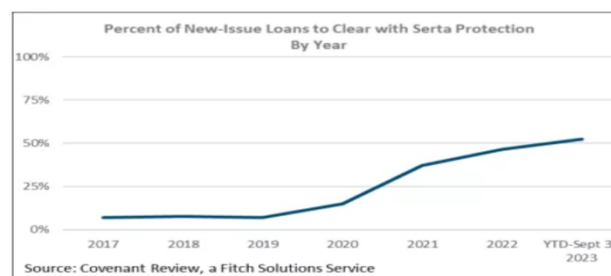
2 How could a non-pro rata uptier be prevented? Have credit agreements widely adopted so-called Serta protection / Serta blockers?

How the Serta “loophole” could be closed was (exhaustively) covered in one of the postscripts within the Serta Postmortem Guide, so I won’t relitigate all of that here.

However, in response to the rise of non-pro rata uptiers, there *has* been a sharp rise in credit agreements that include Serta protection – coming in all shapes and sizes from a wording perspective but most often saying that in order for the liens of existing lenders or their payment priority to be subordinated each adversely affected lender must consent (effectively giving each lender a veto vote on the transaction). But this is often somewhat watered down through saying that if the transaction is open to all lenders to participate in (even if not all lenders *do* participate) then the simple majority consent requirement will be applicable (in other words, a super-priority uptier can occur but everyone must be able to participate in the transaction if they so choose – holders are given the right to participate, but there’s no obligation that everyone must).

Note: No different than with J. Crew, Serta has become the eponym for non-pro rata uptiers even if they’re structured in much different ways (e.g., Incora, Envision, etc.). So, when someone says, “Serta protection” or “Serta blockers” they just mean “non-pro rata uptier protection”.

In response to Serta – and the onslaught of similar transactions that have followed – there has been a market response despite the ongoing litigation over the permissibility of these transactions. Before 2020, Serta blockers were included in under 10% of all newly issued leveraged loans. However, YTD the number of issuances with Serta blockers stands at slightly over 50% and last year was slightly under 50%.



Note: As discussed more below, a valuable non-monetary form of consideration that a company can offer lenders in an amend-and-extend is tightening up the credit agreement to prevent lenders from being caught on the wrong end of a more aggressive transaction in the future. To this end, of the A&Es done over the last few years, Serta protection was in 19% of the docs pre-transaction whereas post-transaction Serta protection was in 49% of them.



Now, I've said before in relation to Serta that it's hard to make an "equity" argument that the transaction should be prohibited because Angelo Gordon, Apollo, et al. were more or less in the wrong place at the wrong time and what happened to them will never happen to anyone else.

If it were the case that there was an onslaught of amendments that occurred in the immediate wake of Serta and that no other non-pro rata transactions *did* happen after Serta – or *could* happen now due to widespread "loophole" closures – then this would be a compelling argument.

However, the fact that around 50% of all new credit agreements have Serta protection of some kind or another may seem to contradict my point a bit (lots of credit agreements *do* now have provisions that preclude this kind of transaction!). But we're three years out from Serta's transaction and, as I wrote in the Serta Postmortem Guide, the insertion of Serta blockers has become a valuable bargaining chip no different than any other in debt doc negotiations. And this reality has helped facilitate A&Es getting done at favorable economic terms in an otherwise tough credit market over the last few years because companies can use a carrot-and-stick approach by not-so-subtly suggesting that if an A&E can't get done then they may need to utilize their more flexible docs to do something more comprehensive (read: aggressive). So, I'd argue (although I'm biased!) that Serta has been a positive for credit markets despite Apollo, et al. suggesting in the immediate wake of the transaction that it'd throw credit markets into a chaotic or catatonic state.

It's also important to keep in mind, as I mentioned above vis-à-vis J. Crew blockers, that looking at the overall loan market can be a bit misleading. Think about it from a healthy company's perspective that's issuing some new secured debt where lenders want the inclusion of Serta protection. The company would need to think about what the value of having the *option* to do a non-pro rata uptier in the future is (even though they can't envision themselves ever in distress) and whether that value is sufficiently high to fight tooth and nail for it in their credit agreement. Now, if the company in question is a PortCo (hint, hint) then the optionality inherent to *not* having Serta protection will be well understood and valued – and this value arises not only from maintaining the *actual* ability to do a non-pro rata uptier, but also from the ability to *threaten* to do one which can allow the sponsor to use a carrot-and-stick approach to get other transactions done as we talked about above vis-à-vis A&Es. So, there's much more of a fight over the inclusion of Serta protections in PortCo credit agreements – and most still don't have them.

However, when it comes to non-sponsor-backed companies in general there's more apathy about this language. So, if lenders want it (as almost all do) then it's much more liable that they'll get it as the company basically thinks, "Well, we'll probably never have to do this kind of transaction anyway so if lenders want this included and we can get some concessions on stuff we care about and that actually matters in the here and now, then sure, fine, whatever, we'll include it."

- 1 Let's imagine that a company files – perhaps due to liquidity running dry – with a \$600mm 1L TL and a \$300mm 2L TL. The debtor is valued at \$1,000mm, it can support around \$500mm of debt post-reorg, and it had a DIP of \$100mm to help it get through its time in-court (assume no roll-up). Given all of this, what's the recovery of the 2L holders and what kind of consideration are they most likely to receive?**

This PWP question is a twist on one of the waterfall questions we talked about last year where we were concerned with both the level of recovery *and* the consideration that would constitute that recovery for various classes. So, a bit of an added level to a classic waterfall where we're concerned with the level of recovery across classes but don't worry ourselves with what, like, that recovery will look like for each class. (Cash? Debt? Warrants? Equity? Some combination?)



So, we know that the pre-filing cap structure consists of the 1L and the 2L. However, post-filing the \$100mm DIP will, as per usual, be in a super-priority position. And, since we've been told there's no [roll-up](#) here, that means we now have a capital structure that includes a \$100mm DIP, a \$600mm 1L, and a \$300mm 2L.

Now, if the enterprise value of the debtor, per the plan, is \$1,000mm then we know everyone is getting a full recovery (100%). However, we also know the *form* of that recovery will have to be different across classes as we've been told the company can only support around \$500mm of debt post-reorg (remember, we were told that they filed due to liquidity issues so maybe this is a tech company that, while its prospects look promising, can't handle too much cash interest now).

There are no hard and fast rules on the composition of recovery across classes. But, as I mentioned in the 2022 Bonus Guide, cash and/or takeback debt will most often be the form of recovery for senior parts of the capital structure (since it's more stable) then post-reorg equity will fill the rest since its value is less stable. This should be somewhat intuitive as one would expect those in a relatively riskier (read: lower) position within the capital structure to receive consideration that is also relatively risky compared to what those more senior in the capital structure receive (even if, as in this case, everyone is getting a full recovery in percentage terms).

So, the most liable outcome here is that the DIP and 1L holders will receive take-back debt (the new \$500mm in debt) or some combination of take-back debt and cash (we'll assume, for the purposes of this question, that they just receive take-back debt here). However, with there only being \$500mm of post-reorg debt contemplated here that means that 1L holders need some other form of consideration to achieve a full recovery (since, after dealing with the DIP, there will only be \$400mm in take-back debt left for the 1L holders and they have \$600mm in total claims).

Therefore, we can say that the consideration received by the 1L holders will likely be \$400mm worth of take-back debt (representing 66.66% of their total recovery) and \$200mm worth of post-reorg equity (representing 33.33% of their total recovery).

Note: Remember that if the debtor, per the plan, has a total enterprise value of \$1,000mm but the plan only contemplates \$500mm of post-reorg debt then the implied post-reorg equity value is \$500mm (or, put another way, the post-reorg equity is worth \$500mm). Therefore, in this case, we're giving 40% of the total post-reorg equity (\$200mm / \$500mm) to the 1L holders. And this amount of post-reorg equity, in conjunction with the take-back debt they're getting, will net them full recovery (100%). (Here we're ignoring any post-petition interest considerations, the amount of pre-filing cash and cash on the balance sheet upon emergence, etc.).

Moving on, we'll assume that the recovery for the 2L holders will come entirely in the form of post-reorg equity. As it happens, since the 1L holders took \$200mm worth of the post-reorg equity, there's \$300mm remaining which mirrors the level of 2L claims outstanding. Therefore, the 2L holders will receive a full recovery (100%) entirely in the form of post-reorg equity which is the equivalent to 60% (\$300mm / \$500mm) of the total post-reorg equity.

As a hypothetical, maybe the reason the company filed was due to the proverbial "good business, bad balance sheet" issue – or, as is more common in rx, the less-proverbial "promising business, bad balance sheet" issue. In other words, the company was promising but simply was suffocated under the weight of too much debt and cash interest (maybe they're a tech company and their balance sheet made sense in the 2020/2021 low rates environment when they were growing quickly, but they've now seen growth fall and their FCF plummet as rates have moved up).



Whatever the case, with a now slimmed down capital structure upon emergence, if the company turns things around and sees its value go up significantly in the years to come, the total return of the 2L holders could be significantly higher than that of the 1L holders – because, obviously, the consideration the 2L holders received (equity) has no capped upside the way the majority of the consideration the 1L holders received (take-back debt) does.

However, the flipside is that if the company emerges and struggles, perhaps seeing its valuation drop a bit (or a lot) because it wasn't quite as promising as some thought, then the 2L holders are fully exposed to that (by dint of their recovery coming fully in equity) whereas the 1L holders are much more insulated by having the take-back debt.

Note: We're assuming, as in all waterfall questions, that there are no administrative claims here (e.g., professional fees for our helpful rx bankers and lawyers) that are super-priority and must be paid in full and in cash for a company to exit bankruptcy.

Note: In the Bonus 2022 Guide I wrote a pretty long digression on all of this (e.g., how more senior parts of the cap stack will, on average, be more liable to receive more “stable” forms of consideration but how there are no hard and fast rules). As another little example, in Lannett's recent prepack (Gugg debtor-side) the 1L holders received their pro-rata share of 97% of the Takeback Exit Facility and 97% of the post-reorg equity for a projected recovery of a little over 90%. Whereas the 2L holders received their pro-rata share of 3% of the Takeback Exit Facility, 3% of the post-reorg equity, *and* some new warrants for a projected recovery of 5%. So, here we see two of the same forms of consideration going to two different classes at much different levels (97% vs. 3%) *plus* a less “stable” form of consideration (warrants) being tossed in for the 2Ls to sweeten the pot a bit more and cajole them to sign off on doing the pre-pack.

Class	Claim/Equity Interest	Treatment of Claim/Equity Interest	Projected Allowed Amount of Claims ⁴	Projected Recovery Under the Plan
3	First Lien Senior Secured Notes Claims	On the Effective Date, except to the extent less favorable treatment is agreed to by the Debtors or the Reorganized Debtors, as applicable, and a Holder of an Allowed First Lien Senior Secured Notes Claim, each Holder of an Allowed First Lien Senior Secured Notes Claim shall receive, in full and final satisfaction of such Allowed First Lien Senior Secured Notes Claim, its Pro Rata share of 97% of the Takeback Exit Facility and its Pro Rata share of 97% of the New Common Stock, subject to dilution on account of the MIP New Common Stock and the New Warrants.	\$364.9 million	91%
4	Second Lien Term Loan Claims	On the Effective Date, except to the extent less favorable treatment is agreed to by the Debtors or the Reorganized Debtors, as applicable, and a Holder of an Allowed Second Lien Term Loan Claim, each Holder of an Allowed Second Lien Term Loan Claim shall receive, in full and final satisfaction of such Allowed Second Lien Term Loan Claim: (1) its Pro Rata Share of 3% of the Takeback Exit Facility; (2) its Pro Rata share of 3% of the New Common Stock, subject to dilution on account of the MIP New Common Stock and the New Warrants; and (3) its Pro Rata share of the New Warrants.	\$224.2 million	5% ⁵

⁵ The projected recovery of Holders of Allowed Second Lien Term Loans excludes any value associated with the New Warrants. As further described in Article IV.B.3(b) herein, the New Warrants are not anticipated to have any value as of the Effective Date.

To illustrate, again, that there are no hard and fast rules here, a little trend over the last few years has been the rise of equity-linked DIP facilities that have resulted in DIP lenders winding up with lots of post-reorg equity (sometimes most of it), those lower in the capital structure getting take-back debt or cash, and those even further down perhaps getting a small sliver of post-reorg equity (or nothing at all depending on how far down the capital structure we go).



There have been a number of ways these facilities have been structured but they boil down to the DIP lenders obtaining the right, but not always the obligation, to convert their DIP loans into equity at some discount to the plan (equity) value or being able to convert on the same terms as are contemplated in the plan's equity rights offering (see my [longer post walking through the economics of a rights offering](#) – all well beyond what you need to know for an interview).

Some recent cases that have involved these include SAS (Apollo up to their old hijinks), Carestream (Apollo in the mix again), Aeromexico (Apollo in the mix... again), Phoenix Services International (Apollo was the sponsor), American Commercial Lines, Avianca, etc. In the end, sure, post-reorg equity is a less “stable” or “certain” form of recovery than cash or take-back debt. But if that post-reorg equity is coming at a discount to the plan (equity) value – which itself a DIP lender might think understates the *real* equity value – then the return structure of getting post-reorg equity is too favorable to not push for. (Some like Carestream involved DIP lenders getting a mix of cash and post-reorg equity as their recovery, so a little of the best of both worlds.)

Note: The Party City RSA caused consternation in some corners earlier in 2023 because it contemplated the DIP lenders (well, the backstop parties) getting the right, but not the obligation, to convert their DIP claims into equity on the same terms as the contemplated equity rights offering. Then the ABL would receive cash or take-back debt and the first-lien notes would receive a relatively small slice of post-reorg equity (after adjusting for the dilution from the DIP conversion, rights offering, and management incentive plan) as their only form of consideration. (Since the DIP backstop parties held over 80% of the first-lien notes it was the minority 1L Noteholders that were most irritated over the RSA as, I mean, the RSA was all structured pre-filing to ensure that upon filing the DIP backstop parties, who held the majority of the 1L, could end up with the majority of the post-reorg equity well below plan value – not a bad outcome if you can finagle it.) (Mudrick threw a fit over all this that lasted for months.)

1 Let's imagine that we're looking at some Senior Secured Notes and we find in the indenture that there's \$100mm available under the credit facilities basket, \$100mm available under the general debt basket, and \$50mm available under the general liens basket. Given this, how much secured debt capacity exists? In other words, how much additional (pari) secured debt will the SSNs allow to be incurred?

In the Double Dip Guide I ventured a bit further into how to think about basket capacity since debt doc constraints inform more or less everything about those transactions (think about all the twists and turns and contortions that Trinseo needed to go through due to debt doc constraints). Plus, I thought it offered a nice excuse for us to walk through a few little examples in more depth. (I also touched on capacity considerations briefly in the Overview Guide and then in a postscript to the Serta Postmortem Guide when discussing how Apollo, et al. were backed into a corner by Serta's credit agreement and thus couldn't offer a much better deal than their preliminary one.)

Now, as I mentioned in the Double Dip Guide, tallying up basket capacity in any meaningful way is well beyond what will be come up in interviews (no one is going to hand over snippets of a 300+ page credit agreement or indenture and ask you to figure out in a few minutes the amount of structurally senior debt that can be incurred!).

However, in a recent EVR *full-time* interview the above question was asked, so I'll cover it and add a few more involved variations below (although it's important to keep in mind that these additional questions are not going to come up – except *maybe* in a lateral context but they haven't before).



As a brief refresher, all debt docs (credit agreements / indentures) are the result of a tug-of-war between the company and its lenders – and the docs will dictate, in a labyrinth of legalese, the *amount* of *additional* debt that can be incurred based on the *priority* it can be incurred at, the *purposes* it can be incurred for, and the type of *entity* that can incur it. (Unless the additional debt that'll be incurred is by an unsub – as we talked about vis-à-vis Serta, Sabre, etc. – because unsubs are outside the restricted group and thus aren't restricted by the negative covenants of the pre-existing debt docs. Therefore, they can incur as much debt as the market will bear – the issue is getting sufficient value to an unsub such that people will *want* to put money to work there since it's not too enticing to lend to an empty shell with no recourse to any value.)

So, the important thing to keep in mind is that if we're looking to incur debt at some restricted group entity (read: one that is bound by the negative covenants of the company's pre-existing debt) we'll be constrained by the pre-existing debt docs and will need to calculate (in dollar terms) how much capacity exists for the specific thing we're trying to do (e.g., incur secured debt, incur unsecured debt, incur structurally senior debt at a non-guarantor restricted sub, etc.).

And this is sometimes much easier said than done because there won't be specific dollar values in the docs that are laid out on a silver platter for us (e.g., the credit agreement or indenture won't say, "We can incur \$200mm more of secured debt, \$400mm of unsecured debt, and \$100mm of debt at a non-guarantor restricted sub"). Instead, the available capacity, or lack thereof, will often be based on a broad set of sources (baskets) within the docs that need to be tallied up together.

Likewise, as we'll discuss later, the description of all the baskets in the docs will be littered with defined terms (e.g., EBITDA, Total Leverage, Fixed Charge Coverage Ratio, etc.) that need to be calculated to *then* calculate the actual dollar amount of capacity available within the basket. However, the definitions of these defined terms within the debt docs can sometimes be a little divorced from reality (sometimes by design, sometimes due to sloppy drafting) and can make putting an exact dollar value on capacity in certain situations, uh, frustrating and/or infuriating.

Note: The restricted group (a.k.a. credit group) simply refers to all the entities in the company's corporate structure that are *restricted* by a company's pre-existing debt docs. So, the restricted group will typically be comprised of the entity (borrower / issuer) that incurred the debt, the subsidiaries that provided a guarantee of the debt, *and* the subsidiaries that didn't guarantee the debt but are nevertheless bound by the negative covenants of the debt docs (non-guarantor restricted subs). In most situations the restricted group will be all (or almost all) of a company's entities except for its unrestricted subs. Now, unrestricted subs, as we talked about before, are outside the restricted group and thus are not restricted by the company's pre-existing debt docs. However, despite how much we talk about unsubs, they're somewhat rare or of little consequence in *most* corporate structures. Most companies want as large of a restricted group as possible because the amount of capacity the borrower / issuer has to incur *additional* debt, issue dividends, etc. will be influenced by the financials of the *entire* restricted group (e.g., for covenant calculation purposes, the EBITDA used will be that of the restricted group and will not typically include the EBITDA contribution of unsubs, for example). So, most companies, in most situations, will forego the more flexible nature of unsubs (read: the ability to incur as much debt as the market will bear) because it makes more sense to have all (or almost all) of the company's entities pooling their assets / earnings to support a few tranches of debt (as opposed to raising debt at a bunch of different entities based on the amount of assets / earnings that exist there). Plus, I mean, debt docs and corporate structures, in a broader sense, are all bargained for and most often lenders will want to lend to the company writ-large for diversification purposes, etc. instead of one or two silos with some specific assets or businesses or whatever located there.



Anyway, with all that said, if you went through the Double-Dip Guide then you may already know the answer to this question. Because one of my mantras – that I repeated over and over and over again – was that if a company was looking to incur additional *secured* debt, or to put in place a *secured* guarantee, the pre-existing debt docs need to provide both debt *and* lien capacity.

So, here we're told that we have a "credit facilities" basket that – to simplify a bit – you can think of as a basket that captures the existing credit facilities and may provide additional (incremental) capacity. Here we haven't had to calculate the capacity available under this basket ourselves: we've simply been told there's \$100mm of capacity available. And, as the name of this basket sort of implies, the credit facilities basket provides both debt *and* lien capacity (there'll be an associated liens basket that pairs with the credit facilities debt basket – in other words, a basket that allows debt raised specifically under the credit facilities debt basket to be secured). So, based on this, we know we can raise at least \$100mm of secured debt by tapping this basket.

Now, think of the general debt basket and the general liens basket as being like two ingredients you need for the recipe of creating secured debt. It's fine to have lots of debt capacity but, since we're interested in figuring out *secured* debt capacity, the binding constraint (in this case) is the smaller amount of lien capacity. So, there's only \$50mm in *secured* debt capacity arising from the general debt and liens baskets (it doesn't matter if we have \$100mm, \$200mm, or \$500mm in the general debt basket, what we're limited by here is the general liens basket of \$50mm).

Therefore, the total amount of additional *secured* debt we can incur here will be \$150mm, with \$100mm stemming from the credit facilities basket and \$50mm stemming from the general debt basket and the general liens basket.

Note: Here's a (terrible) analogy vis-à-vis debt and lien baskets that, if nothing else, will be easy to remember because it's so bad. Imagine someone gives you 100lbs of ground beef (debt capacity) and 10 hamburger buns (lien capacity) and then asks you to make as many hamburgers (*secured* debt) as you can. In the end, the best you can do is make 10 hamburgers even if that results in there being 97lbs of wasted ground beef.

Without getting too much more into the weeds (famous last words) think about the limitation on indebtedness (debt) covenant in an indenture as *prohibiting* additional indebtedness (by any entity within the restricted group) *unless* a ratio test is satisfied on a pro-forma basis (e.g., FCCR > 2x, as that illustrates the company has the EBITDA to service its debt comfortably) in which case unlimited (unsecured) debt can be incurred until that ratio is hit (this is referred to as "ratio debt" for self-evident reasons – however, for this ratio debt to be *secured* there needs to be applicable lien baskets that can be tapped to pair with the ratio debt).

However, even if that ratio test fails – as is the case for many of the companies that we'd care about, and as the question implies here – there are carveouts (baskets) that still allow for additional "permitted indebtedness" such as the credit facilities basket and the general debt basket as we're focused on in this question. In other words, even if we look at the FCCR and realize we're out of luck on the ratio debt front, if we look at the debt baskets we may find significant capacity to incur (unsecured) debt and if we then look to the lien baskets we may find significant capacity to *secure* that debt (although, as in this question, we might find that we're rate-limited on the amount of secured debt we can incur once we look at the size of the lien baskets available).

So, all else equal, a company looking to incur debt will most often opt to max out their ratio debt (if available) before tapping their permitted debt baskets. This, of course, maximizes the total



debt that can be incurred since if a company tapped their debt baskets first then that'll per se reduce the company's FCCR (through raising leverage) and lower the amount of ratio debt available. Therefore, since the permitted debt baskets (e.g., general debt basket) allow for the incurrence of debt even when no ratio debt can be incurred, it makes sense to incur whatever ratio debt one can and then, with the limit hit, move on to the permitted debt baskets as opposed to vice versa. (As we discussed in the Double Dip Guide a bit, there can be a mix-and-matching across baskets to try to incur the total level of debt desired.)

It's also worth briefly touching on reclassification for a second as it'll be mentioned as an aside in a few of the questions I've made up below. In most docs a company will be allowed, from time to time, to reclassify debt that's been incurred through tapping permitted debt baskets (perhaps because there was no ratio debt capacity at the time) as ratio debt and thereby free up capacity in the permitted debt baskets so that they can be used again.

For example, if right now there's no ratio debt capacity (because we aren't in compliance with the ratio test) and we raise \$Xmm through tapping the general debt basket, in the future if FCCR improves and we suddenly now have ratio debt capacity then we can reclassify the debt incurred under the general debt basket as ratio debt and then have the ability to use our general debt basket again.

The major reclassification exception involves the credit facilities (debt) basket because it'll often be sized in line with the company's existing credit facilities *plus* revolver capacity that hasn't been utilized *plus* perhaps some additional headroom for additional secured indebtedness. The historical *raison d'être* of the credit facilities basket was to ensure that even if a company couldn't meet their ratio debt test it could still draw on its revolver (subject to any revolver-specific tests as we touched on earlier).

But because credit facilities make up a sizeable percentage of most capital structures, especially these days, if one could reclassify credit facilities debt as ratio debt, and thus incur more debt under the credit facilities basket, it'd allow for additional *secured* indebtedness that in total could end up being far in excess of what any noteholders likely dreamt of permitting, so that's why there's typically the reclassification exception here (although there are exceptions to this exception in some docs where *some* debt incurred under the credit facilities debt basket can be reclassified as ratio debt).

The credit facilities basket is often the first basket discussed and the longest as there can be a *wide* variance in how one defines credit facilities and the additional incurrence that's permitted under the basket. Whereas the general debt and lien baskets are always more or less a few lines saying, "Here's some debt or lien capacity to do whatever you want with."

Anyway, there are a million caveats here and all debt docs are unique little snowflakes – or at least should be treated as such so you aren't blindsided by small definitional differences (or clever wording changes of typical language) that ends up coming back to bite you. But this is my best attempt at giving you a general introduction to some important principles.

Just remember that the debt covenant will be structured to limit additional indebtedness subject to two exceptions: ratio debt and then a smorgasbord of debt baskets (the latter being able to be tapped even when ratio debt can't be). However, both require you to look elsewhere in the indenture (the lien covenant) to ensure there's applicable lien capacity should you want to incur *secured* debt (or, as discussed in the DD guide, provide a secured guarantee).



- 2 Let's imagine that we're looking at the Senior Secured Notes of ABC Inc. Their entire capital structure consists of an undrawn \$450mm revolver, a TLA of \$500mm, a TLB of \$570mm, and \$570mm of SSNs. Further, their LTM Adj. EBITDA is \$410mm, and the cash and cash equivalents on the balance sheet are \$120mm. Given all this, what's the amount of additional (pari) secured debt the SSNs will allow ABC to incur based on the below baskets found in the indenture?
- **Credit Facilities Basket:** \$1,600mm of debt (all current credit facility debt and all future draws on the revolver must be incurred under this prong and can't be reclassified) *plus* the greater of \$410mm and 100% of EBITDA *plus* pari debt subject to a 3.9x net 1L leverage ratio.
 - **General Debt Basket:** Greater of \$205mm and 50% of EBITDA.
 - **General Liens Basket:** Greater of \$205mm and 50% of EBITDA.

First of all, this will never be asked (or at least should never be because it's more than a bit unreasonable). But I created this to illustrate the fact that in the former question we're being told the dollar amount of capacity available (e.g., there's \$100mm under the credit facilities basket) as if it's some static figure in the indenture. But the capacity available under most baskets will fluctuate based on certain factors (e.g., what EBITDA, as defined in the docs, happens to be or what some leverage ratio, as defined in the docs, happens to be, etc.) and will require a bit of work to figure out.

So, here we have a similar situation to the prior question: we have a credit facilities basket, a general debt basket, and a general liens basket. Therefore, we know that we'll add the amount available under the credit facilities basket to the amount of secured debt capacity we find under the general debt and general lien baskets (e.g., one of these will bottleneck the other unless they happen to be equal to each other as they often are and are here). However, we need to figure out the actual capacity available under each of these baskets given the descriptions we've been given.

It's obvious that we'll have to do the most work (as per usual) in figuring out the credit facilities basket, and that there's some info we need to figure out before getting started. Namely, the amount of applicable secured debt outstanding and the current net 1L leverage ratio (since these will dictate, in part, the size of this basket).

So, the current level of secured debt outstanding that's applicable to the credit facilities basket is the \$500mm TLA and \$570mm TLB that, in total, amounts to \$1.07b. (The revolver would be included if it were drawn but it's not at all, and the SSNs are obviously secured debt but are unlikely to fall under the "credit facilities" definition used in the indenture – although always look to see how defined terms are defined in debt docs as the definitions can and will surprise you!)

Therefore, if we move through the credit facilities basket step-by-step, we'll have: \$530mm available (\$1,600mm - \$500mm - \$570mm) *plus* ii) the greater of \$410mm and 100% of EBITDA (\$410mm), which will be \$410mm *plus* iii) the amount of 1L debt that makes us hit the 3.9x net 1L leverage from our *current* starting point. So, since our current net 1L leverage is $(\$500\text{mm} + \$570\text{mm} + \$570\text{mm} - \$120\text{mm}) / (\$410\text{mm}) = 3.7073\text{x}$ the amount available up to 3.9x is $(3.9 - 3.7073) * (\$410\text{mm}) = \79mm . Putting all the pieces of the puzzle together, we'll have \$530mm + \$410mm + \$79mm = \$1,019mm available under the credit facilities basket.

Moving on to the general debt basket and the general liens basket, as is often the case these are laid out the same. Therefore, the amount available under each is \$205mm and the greater of 50% of EBITDA ($\$410\text{mm} * 0.5 = \205mm).



So, the amount of *secured* debt capacity available is \$205mm – with, once again, the *debt* capacity coming from the general debt basket, the *lien* capacity coming from the general liens basket, and both being necessary ingredients to make the *secured* debt we’re looking for here.

Given all this, if we add up the capacity under the credit facilities basket and the general debt / liens baskets, the total amount of *pari secured* debt we can incur is $\$1.019\text{mm} + \$205\text{mm} = \$1,224\text{mm}$. So, the pro-forma net leverage, assuming we took advantage of all the *secured* capacity, would be $(\$500\text{mm} + \$570\text{mm} + \$570\text{mm} + \$1,124\text{mm} - \$120\text{mm}) / (\$410\text{mm}) = 6.7\text{x}$.

Here’s an example of how all of this is worded in the wild – in a labyrinth of legalese. First, the debt covenant starts off, as per usual, with a bit of a whiplash by saying, “The company can’t incur additional debt – however, I mean, it’s fine to incur more debt if that keeps the company in compliance with one or two ratios.” So, here we see that additional *unsecured* debt (ratio debt) can be incurred if the FCCR (as defined in the docs) is no lower than 2.0x or the Total Leverage Ratio (as defined in the docs) is no greater than 4.40x after incurring this new debt (read: if pro forma we’re still in compliance with these ratios).

Note: To be clear, we’re ignoring ratio debt considerations in this question. I’m just putting this here to show the general flow of an indenture and to tie it back to what we talked about in the prior question. Also, keep in mind that for any hypothetical ratio debt to be *secured* we’d need there to be applicable *lien* capacity (if any exists) found elsewhere in the indenture.

Section 4.09 Limitation on Indebtedness.

(a) The Company shall not, and shall not permit any of its Restricted Subsidiaries to, directly or indirectly, incur any Indebtedness; *provided, however*, that the Company and any Restricted Subsidiary may incur Indebtedness if on the date of such incurrence and after giving Pro Forma Effect to the incurrence of such Indebtedness and to any Specified Transaction to be consummated in connection therewith, for the most recent Test Period, either (A) the Fixed Charge Coverage Ratio is no lower than (i) 2.00:1.00, or (ii) in the case of Indebtedness incurred in connection with Permitted Acquisitions or similar Permitted Investments (other than those described in clause (32) of the definition of “Permitted Investments”), the Fixed Charge Coverage Ratio immediately prior to the incurrence of such Indebtedness and the consummation of such Acquisition or other Permitted Investment, or (B) the Total Leverage Ratio is no greater than (i) 4.40:1.00, or (ii) in the case of Indebtedness incurred in connection with Permitted Acquisitions or similar Permitted Investments (other than those described in clause (32) of the definition of “Permitted Investments”), the Total Leverage Ratio immediately prior to the incurrence of such Indebtedness and the consummation of such Acquisition or other Permitted Investment (recomputed for the foregoing clauses (A) and (B) for the applicable Test Period); *provided* that the aggregate principal amount of all Indebtedness incurred and outstanding under this Section 4.09(a) of Non-Guarantor Subsidiaries, when aggregated with the aggregate principal amount of all other Indebtedness incurred by Non-Guarantor Subsidiaries then outstanding pursuant to Section 4.09(b)(1), after giving Pro Forma Effect to such incurrence and other transactions and the use of the proceeds thereof, shall not exceed the greater of (x) \$185.0 million and (y) 45.0% of Consolidated EBITDA for the most recent Test Period.

Note: Often, as above, there’ll be a “sublimit” on the amount of ratio debt that can be incurred by non-guarantor restricted subs – since debt incurred at non-guarantor restricted subs, as I talked about in the DD Guide, will have a structurally senior claim on the assets residing within those subs relative to the pre-existing debt (the cap here is the greater of \$185mm and 45% of EBITDA).

Then, right after this we start with the classic baskets. Notice how this begins with a sentence basically saying, “Despite what was said earlier in Section 4.09(a), the following are exceptions that allow for additional debt to be incurred. In other words, these are little carve-outs where Section 4.09(a) doesn’t limit what can be done even if we’re not in compliance with the ratios outlined above right now.”

Below is the credit facilities (debt) basket and I’ve also included the associated liens basket that’s found elsewhere in the indenture (within the liens covenant)...



(b) Section 4.09(a) shall not prohibit the incurrence of the following Indebtedness:

(1) Indebtedness of the Company or any Restricted Subsidiary incurred under a Credit Facility (including the Senior Credit Facility), in an aggregate amount at any time outstanding up to (a) the sum of (x) \$1,600.0 million and (y) the greater of (A) \$410.0 million and (B) 100% of Consolidated EBITDA for the most recent Test Period plus (b) an unlimited amount if, after giving Pro Forma Effect to the incurrence of such Indebtedness and to any Specified Transaction to be consummated in connection therewith, for the most recent Test Period, (x) in the case of Indebtedness that is secured by a lien on the Collateral that is *pari passu* with the liens securing the Notes, the First Lien Leverage Ratio is no greater than (i) 3.90:1.00, or (ii) in the case of Indebtedness incurred in connection with Permitted Acquisitions or similar Permitted Investments (other than those described in clause (32) of the definition of “Permitted Investments”), the First Lien Leverage Ratio immediately prior to the incurrence of such Indebtedness and the consummation of such Acquisition or other Permitted Investment and (y) in the case of Indebtedness that is secured by a Lien on the Collateral that is junior to the liens securing the Notes, the Secured Leverage Ratio is no greater than (i) 4.15:1.00, or (ii) in the case of Indebtedness incurred in connection with Permitted Acquisitions or similar Permitted Investments (other than those described in clause (32) of the definition of “Permitted Investments”), the Secured Leverage Ratio immediately prior to the incurrence of such Indebtedness and the consummation of such Acquisition or other Permitted Investment (recomputed for the foregoing clauses (b)(x) and (b)(y) for the applicable Test Period); *provided* that the aggregate principal amount of all Indebtedness incurred and outstanding under this clause (1) of Non-Guarantor Subsidiaries, when aggregated with the aggregate principal amount of all other Indebtedness incurred by Non-Guarantor Subsidiaries then outstanding pursuant to Section 4.09(a), after giving Pro Forma Effect to such incurrence and other transactions and the use of the proceeds thereof, shall not exceed the greater of (x) \$185.0 million and (y) 45.0% of Consolidated EBITDA for the most recent Test Period;

“Permitted Liens” means, with respect to any Person:

(1) Liens securing Indebtedness and other obligations of the Company and its Restricted Subsidiaries under a Credit Facility permitted to be incurred under Section 4.09(b)(1) and having such ranking as set forth therein; *provided* that a senior representative acting on behalf of the holders of such Indebtedness shall have entered into (or otherwise become party to) (A) if such Indebtedness is secured by a Lien on the Collateral that is *pari passu* (but without regard to the control of remedies) with the Liens securing the Notes, a Customary Intercreditor Agreement with the Collateral Agent which agreement shall provide that the Liens on the Collateral securing such Indebtedness shall be secured by the Collateral on a *pari passu* basis with the Liens on the Collateral securing the Notes and (B) if such Indebtedness is secured by a Lien on the Collateral that is junior to the Liens securing the Notes, a Customary Intercreditor Agreement with the Collateral Agent which agreement shall provide that the Liens on the Collateral securing such Indebtedness shall rank junior in priority to the Liens on the Collateral securing the Notes; *provided, further*, that without any further consent of the Holders, the Collateral Agent shall be authorized to execute and deliver on behalf of the Notes Secured Parties any Customary Intercreditor Agreement or any amendment (or amendment and restatement) to the Collateral Documents or a Customary Intercreditor Agreement to the extent necessary to effect the provisions contemplated by this clause (1);

And here are the general debt and general liens baskets – much easier on the eyes...

(22) Indebtedness of the Company and its Restricted Subsidiaries not exceeding in the aggregate at any time outstanding the greater of (x) \$205.0 million and (y) 50.0% of Consolidated EBITDA for the most recent Test Period;

(21) Liens securing Indebtedness and other obligations in an aggregate principal amount at any one time outstanding not to exceed the greater of \$205.0 million and 50.0% of Consolidated EBITDA (determined at the time of incurrence thereof for the most recent Test Period);

Note: It’s a challenge to try to condense this all down, so all this basket talk might seem as clear as mud right now. However, with your vocabulary built up a bit (even if it doesn’t seem like it!) it’ll now be much easier to peck through more exhaustive debt doc overviews and recognize what’s important to pay attention to. To this end, for a more comprehensive treatment on parsing *both* indentures and credit agreements see the [excellent overview from Simpson Thacher](#) (brief [indenture-only version](#) here) or the more basic indenture overviews [from Covenant Review](#) or [from Paul Weiss](#). (There are some differences in terms and structure between credit agreements and indentures, but the foundation is all similar). But bear in mind that a lot of what’s contained in an indenture or credit agreement – and that’s talked about in the above overviews – isn’t too important to us most of the time (e.g., reporting stuff, etc.). What’s most important to look at (most of the time) will be the debt covenant, lien covenant, and restricted payments covenant. Because these then allow us to figure out the amount of secured or unsecured debt that can be



incurred, the amount of structurally senior debt that can be incurred, the amount that can be shifted to an unsub, and the amount that can flow out of the restricted group more generally (e.g., for dividends). However, this is *not* important for interviews and is *light years* beyond what'll ever come up. This will all be learned by osmosis and isn't worth spinning your tires on beforehand. But since I decided to venture down this rabbit hole more with these questions, I figured I'd make them more realistic to at least build up your vocabulary and give you a rough sense of things.

1 Let's imagine that we're looking again at the Senior Secured Notes of ABC Inc. As a reminder their LTM EBITDA is \$410mm, cash and cash equivalents are \$120mm, and the total amount of debt inclusive of these SSNs is \$1,640mm. Based on the below baskets in the indenture, how much can be transferred to an unrestricted subsidiary?

- **Credit Facilities Basket:** \$1,600mm of debt (all current secured debt and all future draws on the revolver must be incurred under this prong and can't be reclassified) *plus* the greater of \$410mm and 100% of EBITDA *plus* pari debt subject to a 3.9x net 1L leverage ratio.
- **General Investments Basket:** Greater of \$240mm and 60% of EBITDA.
- **Investments in Unrestricted Subsidiaries:** Greater of \$105mm and 25% of EBITDA.
- **Investments in Similar Business:** Greater of \$120mm and 30% of EBITDA.
- **Investments in Joint Ventures:** Greater of \$200mm and 50% of EBITDA.
- **General Restricted Payments:** Greater of \$145mm and 35% of EBITDA.
- **Builder Basket (2x FCCR or 3.4x net total debt ratio test for access to CNI component):** Based on 50% of Cumulative Net Income (CNI) from issuance *plus* the greater of \$105mm and 25% of EBITDA (assume CNI is \$0mm).
- **Leverage-Based Investments:** Amounts in compliance with a Total Leverage Ratio of 3.4x.

Once again, this isn't something you have to worry about. But, as I discussed in the Serta Postmortem Guide and in relation to Trinseo, there are certain baskets that one will look to in figuring out how much value can be transferred down to an unrestricted sub.

Sure, all credit agreements and indentures are unique snowflakes with their own unique mix of baskets with their own unique limitations. However, the tell-tale sign of the baskets that *can* be used to transfer value outside the restricted group (e.g., to do a drop-down transaction) is that they'll have the word "investments" in them. Then others that can often be used are ones like the general restricted payments basket and builder (a.k.a. buildup) basket (these latter two, along with others like the leverage-based restricted payments basket which can be tapped so long as a certain leverage ratio is maintained, can alternatively be tapped to make regular-way dividends).

The easiest way to think about all of these baskets is that they're all carveouts that can allow a certain amount of value to flow out from the restricted group for certain purposes – some broad and some more narrow depending on how they're defined in the docs. And this amount of value – in most debt docs, for most of the above baskets – will increase as the company performs better (as most are structured as the greater of a flat amount and some percent of EBITDA or total assets or whatever if this latter amount happens to be higher – therefore, as EBITDA or total assets or whatever rises so too does the amount that can flow out from the restricted group).

So, in this situation the credit facilities basket isn't applicable as it allows us to incur additional debt at a secured level *within* the restricted group, and what we're trying to assess is how much value we can flow *out* to an unsub. However, the rest of the baskets I've listed are applicable – so, let's tally up the capacity and take a look at how these baskets are worded in one indenture...



There's \$246mm ($\$410\text{mm} \times 0.6$) of capacity under the general investments basket, that's defined here as...

(13) Investments not exceeding the greater of \$240.0 million and 60.0% of Consolidated EBITDA (determined at the time such Investment is made for the most recent Test Period and valued at the time of the making thereof, and without giving effect to any write-downs or write-offs thereof) at any time outstanding (net of any return in respect thereof, including dividends, interest, distributions, returns of principal, profits on sale, repayments, income and similar amounts);

There's \$105mm (as that's a bit larger than $\$410\text{mm} \times 0.25 = \102.5mm) of capacity under the investments in unrestricted subsidiaries basket, that's defined here as...

"Permitted Investments" means:

(1) Investments (i) by the Company or a Subsidiary Guarantor to or in the Company or any other Subsidiary Guarantor, (ii) by the Company or any Restricted Subsidiary in any Unrestricted Subsidiary in an aggregate amount at any time outstanding not to exceed the greater of \$105.0 million and 25.0% of Consolidated EBITDA (determined at the time such Investment is made for the most recent Test Period), (iii) by the Company or any Subsidiary Guarantor to or in any Non-Guarantor Subsidiaries, (iv) by a Non-Guarantor Subsidiary to or in another Non-Guarantor Subsidiary, the Company or any Subsidiary Guarantor and (v) by the Company or any Subsidiary of the Company to or in the Company or any Subsidiary of the Company so long as such Investments are part of a series of substantially concurrent transactions that result in the proceeds of such Investments ultimately being invested in (or distributed to) the Company or any Restricted Subsidiary;

There's \$123mm ($\$410\text{mm} \times 0.30$) of capacity under the investments in similar business basket, that's defined as...

(17) Investments in a Similar Business having an aggregate fair market value, taken together with all other Investments made pursuant to this clause (17) to the extent outstanding, not to exceed the greater of \$120.0 million and 30.0% of Consolidated EBITDA (determined at the time such Investment is made for the most recent Test Period and with the fair market value of each Investment being measured at the time made and without giving effect to subsequent changes in value); *provided, however*, that if any Investment pursuant to this clause (17) is made in any Person that is not a Restricted Subsidiary at the date of the making of such Investment and such Person becomes a Restricted Subsidiary on or after such date, such investment shall thereafter be deemed to have been made pursuant to the applicable provisions of clause (1) above and shall cease to have been made pursuant to this clause (17) for so long as such Person continues to be a Restricted Subsidiary;

There's \$205mm ($\$410\text{mm} \times 0.50$) of capacity under the investments in joint ventures basket (if JV isn't strictly defined, can interpret it broadly to use in unsub transfers), that's defined as...

(16) Investments in joint ventures, partnerships and the like in the aggregate at any time outstanding not to exceed the greater of \$200.0 million and 50.0% of Consolidated EBITDA (determined at the time such Investment is made for the most recent Test Period);

There's \$145mm (as that's a bit larger than $\$410\text{mm} \times 0.35 = \143.5) of capacity under the general restricted payments basket, that's defined as...

(11) the Company or any of its Restricted Subsidiaries may make Restricted Payments in an aggregate amount not to exceed the greater of \$145.0 million and 35.0% of Consolidated EBITDA (determined at the time such Restricted Payment is declared (if such Restricted Payment is in the form of a dividend) or is made (in the case of any other Restricted Payment) for the most recent Test Period) if no Event of Default specified in Section 6.01(a)(1), (2), (8) or (9) shall have occurred and be continuing;

There's \$105mm (as that's a bit larger than $\$410\text{mm} \times 0.25 = \102.5mm) of capacity under the builder basket (we'll assume there's no CNI buildup, as it's based on the time since issuance and we'll just pretend that these SSNs were just issued) that's defined as...



(b) with respect to any Restricted Payment made in reliance on clause (c)(i)(b) below of this Section 4.07(a), either (1) the Total Leverage Ratio, calculated on a Pro Forma Basis as of the most recent Test Period prior to the date of declaration, is equal to or less than 3.40:1.00 or (2) the Fixed Charge Coverage Ratio, calculated on a Pro Forma Basis as of the most recent Test Period prior to the date of declaration, is no lower than 2.00:1.00; and

(c) the aggregate amount of such Restricted Payment and all other Restricted Payments declared or made subsequent to the Issue Date in reliance on this clause (c) (directly or indirectly via Section 4.07(b)(5) in respect of dividends or other distributions on Capital Stock) would not exceed at any time of determination, an amount not less than zero in the aggregate, determined on a cumulative basis, equal to, without duplication, the sum of:

(i) (a) the greater of (x) \$105.0 million and (y) 25.0% of Consolidated EBITDA for the most recent Test Period plus (b) an amount equal to 50.0% of Consolidated Net Income for the cumulative period from April 1, 2023 to and including the last day of the most recent Test Period;

There's \$0mm of capacity under the leverage-based investments basket, as the Total Leverage Ratio (which, despite what the name suggests, is defined as net of unrestricted cash in the indenture) is 3.7x which is, of course, higher than the 3.4x test here...

(20) Investments to the extent that, upon giving Pro Forma Effect to the making of such Investment and any Specified Transaction to be consummated in connection therewith, as of the last day of the most recent Test Period, the Total Leverage Ratio is not greater than 3.40:1.00;

So, if we add all of this up we find that there's \$929mm in value that can be transferred to an unrestricted sub right now (could be higher in the future depending on how the company does!).

Note: I've said many times over the years that when looking at debt docs, defined terms mean what they're defined to mean within the docs – not what one thinks they ought to mean or should mean or have meant in other docs. The language is all bargained for – and this often means the definition of terms such as EBITDA that'll be used in calculating capacity is a bit removed from what one would think a company's actual EBITDA is (the EBITDA definition can stretch for pages with lots of ambiguous add-backs, some of which are capped to provide a bit of upper-bound certainty to lenders). Likewise, the way that ratios are calculated (e.g., the Total Leverage Ratio above) or even what constitutes "debt" can also be a bit (or a lot) different than expected. But these definitions aren't meant to be the most accurate representation of the company's economic reality – they're simply bargained for definitions that are the end result of borrower's mostly trying to maximize available capacity and lenders trying to mostly restrict it.

1 Let's imagine that we're trying to figure out how much structurally senior debt (read: debt at some non-guarantor restricted sub) that XYZ Inc. can incur. Based on the below baskets outlined in the credit agreement of XYZ's Term Loan, what do you think? (Per XYZ's most recent financials, total assets on the balance sheet are \$2,000mm).

- **General Debt Basket:** Greater of \$80mm and 5.0% of total assets.
- **General Liens Basket:** Greater of \$40mm and 2.5% of total assets.
- **Non-Guarantor Debt Basket:** Greater of \$80mm and 5.0% of total assets.
- **Non-Guarantor Liens Basket:** Permits non-guarantor debt to be secured by available non-guarantor assets.

We talked about this situation vis-à-vis Trinseo because even though it appears that the general liens basket will constrain the amount of the general debt basket we can use, this isn't actually the case because, so long as we're incurring debt at a non-guarantor restricted sub, the non-guarantor liens basket provides us the lien capacity we need to pair with the debt capacity coming from *both* the general debt basket and the non-guarantor debt basket (assuming there's a sufficient amount of assets at the non-guarantor subs to secure the debt against!).



So, here we'll have the ability to incur $\$100\text{mm} + \$100\text{mm} = \$200\text{mm}$ of secured debt at a non-guarantor restricted sub – thus providing holders of this debt a structurally senior claim on the assets residing there. (Remember that 5.0% of total assets is $\$100\text{mm}$, so that's why we're using $\$100\text{mm}$ for the general debt basket and non-guarantor debt basket not $\$80\text{mm}$ for each.)

Note: As a little aside, in the Double Dip Guide I mentioned that with rare exception guarantees are considered the equivalent to incurring new debt and thus require debt and lien capacity too (the latter being necessary if it's a secured guarantee). Therefore, in this situation, debt that was issued out of some other entity could be provided a secured guarantee (capped at $\$200\text{mm}$) from the company's non-guarantor subs similar to what we talked about vis-à-vis Sabre and Trinseo (this secured guarantee would provide lenders a structurally senior claim on the assets residing within the non-guarantor subs relative to the pre-existing debt).

1 Imagine that a company has EV / EBITDA of 10x and Debt / EBITDA of 4x. Assuming the company has a market cap of \$360mm, how much debt does it have?

Similar to the classic rx question where you're given a leverage ratio and coverage ratio and then need to find the yield on the debt, this question is closer to a brainteaser than a question that actually, like, tests your rx knowledge.

The best thing to do in a situation like this is express the multiples in common terms if there's some shared variable. So, in this case we can just say that $\text{EV} = 10(\text{EBITDA})$ and $\text{Debt} = 4(\text{EBITDA})$. Now, we know that EV, in this simple scenario, will be $\text{EV} = \text{Debt} + \text{Market Cap}$, and we've been told what the market cap is. Therefore, we can express the EV and Debt components of this question in common terms by turning $\text{EV} = \text{Debt} + \360mm into $10(\text{EBITDA}) = 4(\text{EBITDA}) + \360mm or, rearranging a bit, $6(\text{EBITDA}) = \$360\text{mm}$. So, our EBITDA will be $\$360\text{mm} / 6 = \60mm .

Plugging $\$60\text{mm}$ into our slightly rearranged formulas above will give us an EV of $\$600\text{mm}$ and Debt of $\$240\text{mm}$. And, if we check the original formulas in the question, we find that $\$600\text{mm} / \$60\text{mm} = 10\text{x}$ and $\$240\text{mm} / \$60\text{mm} = 4\text{x}$.

1 Let's imagine that ABC Inc. has a HoldCo that's issued \$100mm of Senior Notes and holds 100% of OpCo1's equity and 100% of OpCo2's equity. Further, let's imagine that HoldCo sent part of the proceeds from the Senior Notes to OpCo1 in the form of a \$40mm (unsecured) intercompany loan and, in return, received an intercompany loan receivable. Today, HoldCo has no other assets beyond OpCo1 equity, OpCo2 equity, and the intercompany loan receivable; OpCo1 has an EV of \$50mm, Senior Secured Notes of \$30mm, and Senior Notes of \$10mm; and OpCo2 has an EV of \$80mm, Senior Secured Notes of \$40mm, and Senior Notes of \$20mm. Given all this, what are the recoveries here assuming that there are no guarantees?

So, since we know that the recovery of HoldCo's SNs will probably be influenced by the residual equity value flowing from one of the OpCos, let's start with one of them. For OpCo1, we have an EV of $\$50\text{mm}$ and, after dealing with the $\$30\text{mm}$ of SSNs that'll receive full (100%) recovery, we have $\$20\text{mm}$ in distributable value left against $\$50\text{mm}$ of unsecured claims (the $\$10\text{mm}$ of SNs and the unsecured $\$40\text{mm}$ intercompany loan). Therefore, the recovery for both will be 40% or $\$4\text{mm}$ for the SNs and $\$16\text{mm}$ for the intercompany loan. And, of course, there'll be no residual equity value that'll flow to HoldCo.



For OpCo2, we have an EV of \$80mm and that more than covers both the \$40mm of SSNs and the \$20mm of SNs, so both receive a 100% recovery. This leaves \$20mm of value that'll then flow to HoldCo since HoldCo holds 100% of OpCo2's equity.

Given this, if we think about HoldCo there are \$100mm of SNs outstanding and two sources of value: the \$20mm from OpCo2, and the \$16mm from the intercompany loan. Since remember that the intercompany loan was made *from HoldCo* to the *benefit of OpCo1* – so the recovery of the intercompany loan we determined in OpCo1's waterfall is recovery that "belongs" to HoldCo, and we haven't yet determined what *creditors* will benefit from this intercompany loan recovery. So, the waterfall at HoldCo is straight-forward: there are \$100mm of SNs and there's \$36mm of distributable value, so HoldCo's SNs will receive a 36% recovery.

Put another way, if we think about HoldCo it has \$100mm of liabilities (the SNs) and three assets: 100% of OpCo1 equity (that we've determined is worth \$0mm), 100% of OpCo2 equity (that we've determined is worth \$20mm), and a \$40mm intercompany loan receivable (that we've determined is worth \$16mm). Therefore, we have \$36mm in value against \$100mm of claims, thus the 36% recovery for HoldCo's Senior Notes.

Note: This kind of question hasn't been asked before (although EVR asks a similar one sans the intercompany loan component). But I figured I'd add it since it's a simpler interview-style waterfall question that involves an intercompany loan receivable relative to the (real-world) waterfall involving an intercompany loan receivable that I went through in the Double-Dip Guide vis-à-vis Wheel Pros that's a bit more complicated.

- 1 **Imagine that XYZ Inc. has a pretty simple capital structure consisting of an undrawn \$135mm ABL maturing in July of 2027, a \$500mm TL at S + 4.0% (SOFR is 5.0%) maturing in October 2025, and \$270mm of Senior Notes at 7.75% maturing in October of 2026. Now let's imagine that the aim here is to do an amend-and-extend of the inside maturity (the TL). How could we try to craft something that the TL holders would be interested in assuming that there's \$40mm of cash on the balance sheet and levered FCF has been \$1-3mm per quarter over the last year?**

Remember that in a restructuring context an amend-and-extend will be done in a situation where there's some piece of debt maturing in the not-too-distant future *but* the company isn't too bullish on the prospect of being able to do a regular-way refi of this debt. So, an amend-and-extend is a way to buy the company a bit more time to right the ship so that it can properly address its maturity wall(s) in the future – or at least figure out how to do a more comprehensive restructuring transaction in the future.

However, it's important to keep in mind the incentive structure here. If you're a secured creditor maybe you look at the company's current financials and think to yourself, "This business is slowly falling apart but because of where we reside in the capital structure we'll still be fully covered – or at least will have a pretty high recovery – if the company files any time before our debt matures. However, if we agree to an extension and the company's value falls off a cliff in that *extension period*, then we could end up being heavily impaired. So, to compensate for the inherent risk of exposing ourselves to this company for even longer we need something worthwhile in return."

Plus, I mean, if the company is desperate to do an amend-and-extend and you're holding the debt they want to extend then you have some obvious leverage here (although the leverage in an A&E isn't completely lopsided toward holders as we'll discuss vis-à-vis the docs).



Anyway, in an amend-and-extend there are three main sources of value that most often will be utilized to cajole holders to consent to an extension: two monetary, one non-monetary. First, a cash paydown. Second, an increase in rate. Third, a tightening of the docs.

So, in this example, there's not too much cash on the balance sheet and FCF is barely positive (in other words, there's no cash burn happening right now but they're pretty close to the precipice). Given this, the company doesn't have the ability to offer a huge cash paydown (e.g., paydown \$100mm of the TL at par) or offer a big bump in the rate (e.g., offer S + 700 since that could push them into cash burn territory). Although, needless to say, either or both of these would probably cajole the TL holders to extend all else equal. However, a nice compromise that doesn't wreck the company's financials could be enhancing the TL economics by 100bps to S + 5.0% and doing a little cash paydown of 4% (\$20mm) using cash on the balance sheet.

Now, these two alone may not be enough to incentivize the TL holders to extend. But the third (non-monetary) form of consideration in conjunction with these two could be. Because chances are this company, like most others these days, has a pretty loose credit agreement that would allow for some more creative transactions to occur that could harm the value of the TL – or at least make its future prospects a bit shakier than right now.

So, in addition to the modest economic enhancements, the company could offer to tighten the credit agreement by, for example, limiting the level of additional 1L incurrence (in other words, decreasing or removing the possibility of incurring more pari debt via some of the baskets we've discussed above), inserting a Serta blocker, inserting a J. Crew blocker, limiting the amount that can be incurred at non-guarantor subs, and/or limiting the amount that can be transferred to unrestricted subs. Basically, ensuring that not too much additional debt can be placed pari or senior to the TL and ensuring that not too much value backing the TL can leak away from them.

This is the reason I mentioned before that not all the leverage is per se in the hands of the TL holders here (even though the company may *really* need to extend out the maturity soon) because the company can basically say, "It's no secret that we're in a tough spot – but we're bullish on our prospects due to our incredible cost-cutting and growth initiatives that we're going to roll out over the next few years (*wait till you see 'em*). However, when you're in a tough spot, tough choices need to be made. So, if we can't extend out this TL we'll need to explore some more comprehensive transactions to buy ourselves time – perhaps through a LME, and no we won't be more specific and say what kind of LME because we don't really know what it means – to ensure we have the runway to see our turnaround through to fruition."

The above is a bit tongue-in-cheek. But there's a carrot-and-stick approach here, as is the case for lots of A&Es these days: offering a nice little package to entice holders to extend, but then not-so-subtly implying that if it's not taken then holders might not like what comes next.

Now, as a final note, we've covered the "amend" part but haven't talked much about the "extend" part – in other words, *when* the TL would be extended to. Given the setup here, we could bump it out a few years (it doesn't really matter) but the TL holders would demand that there be a springer such that *if* there are still Senior Notes outstanding three months prior to their maturity (Oct '26) then the TL would spring forward to that date (April '26) so that they remain the inside maturity.

So, this wouldn't buy a whole lot of time if the Senior Notes can't be dealt with down the road, but the idea here would be that the company would feel more comfortable trying to figure out a way to deal with these Senior Notes later on when they're in a stronger (hopefully!) financial situation



(although because of the tightened credit agreement as part of this hypothetical A&E, there are going to be a more limited set of options for addressing those SNs when the time comes!).

Note: This is a longer answer that I anticipated giving here. Needless to say, it's all a bit more than you need to know and much more than you need to regurgitate. However, the main thing I wanted to get across was how tightening the docs *is* a valuable form of consideration to give – and, from the company's perspective, one that doesn't come attached with a cash cost but that can limit the other restructuring options that are available to them down the road.

Note: The specifics of how the credit agreement would be tightened will be contingent on its structure and current level of permissibility – but I've covered the main things above that would be looked to. However, in an ideal world the company will offer to tighten *certain* areas of the credit agreement but keep some looseness elsewhere to ensure they still have some options (albeit a narrower set of them) down the road if a regular-way refi won't work and they need to do another transaction to bring in some more liquidity, deal with the SNs in 2026, etc.

- 1 **Let's imagine that XYZ's Term Loan is trading around 75 with a YTM of around 25%. The little A&E that's being mulled over now would involve XYZ using cash on the balance sheet to paydown 4% of the TL at par, enhance the TL's economics by 100bps to S + 500bps, and tighten up the credit agreement (e.g., decrease additional 1L incurrence, add J. Crew and Serta blockers, etc.). All in return for the TL maturity being bumped out a few years. If we believe that as a result of this transaction the TL's YTM will tighten to 17.0% – implying a post-transaction price of around 80 – how could we think about a TL holder's "return" from the transaction?**

So, here we're thinking that if the transaction occurs the YTM will tighten (fall) a bit because we pushed out a maturity and, if we assume the company's FCF situation doesn't become too dire, the probability of filing has been lowered. Plus, if we assume the credit agreement was pretty loose beforehand, through tightening up the credit agreement the possibility has been diminished that the company will do something that could eviscerate the position / recovery of the pre-existing holders through a more creative transaction (e.g., a non-pro rata uptier or drop-down) or just threaten their recovery through lots of additional 1L incurrence that would reside pari to them (so, this doc tightening would be a net positive for pricing as it reduces downside risk to holders).

Needless to say, the pro-forma YTM is a guesstimate and you'd want to look at how the transaction would look at various pro-forma levels (e.g., 16%, 18%, 20%, etc.). However, if we assume that the YTM will tighten to 17%, and that the TL trades at 80 after the transaction, we can figure out what the immediate "return" would look for holders (of course, most will have bought the TL at much higher levels so it's not like their real return will suddenly be positive by dint of the transaction – the "return" here should be thought of as what a holder has "made" as a direct consequence of the transaction being done).

So, first, we know pre-transaction that there's \$500mm of the TL outstanding and it's trading at 75. Therefore, the market value of the TL is \$375mm. Second, we know holders will receive their pro-rata share of a 4% paydown of the face value of the TL, in cash, that amounts to \$20mm in total (\$500mm * 0.04). Third, we know the new market value of the TL will be the pro-forma amount of debt (\$480mm due to the \$20mm paydown) and we think that the TL will trade around 80 post-transaction. Therefore, the market value of the TL will be \$382 (higher than before, even with the lower face value amount!).



So, we can think about the return to holders as being the \$20mm in cash *plus* the new market value of the debt that they still hold *minus* the current market value of the TL, or \$20mm + \$382mm - \$375mm = \$28mm. Thus implying an investor pickup (return) of 7.5% (\$28mm / \$375mm) due to the transaction.

	Debt Amount	All-In Rate	Maturity	Market Price	Implied YTM	Market Value
Current	\$500	9.00%	Oct-25	75	24.8%	\$375
	Debt Amount	Paydown %	Value of Cash	Paydown \$		
Paydown	\$500	4%	100%	\$20		
	PF Debt Amount	New All-In Rate	New Maturity	Implied Price ¹	Market Value	
Extension	\$480	10.00%	Oct-27	\$80	\$382	

¹ Assumes that TL YTM tightens to 17.0% post-transaction based on the lower probability of filing, longer duration, and tighter docs that preclude additional 1L incurrence, value leakage, etc. that could diminish recoveries in the future.

Cash Paydown Value	\$20
Maturity Extension Value	382
Total Transaction Value	\$402
Less: Current Market Value of TL	(375)
Total Investor Pickup (\$)	\$28
Total Investor Pickup (%)	7.5%

2 If collateral is stripped from pre-existing secured lenders (e.g., through a drop-down) what are some ramifications for these lenders if the company files down the road and their collateral value is now less than their total claim?

This specific question would never come up, but I figured I'd jot down a few quick notes here...

First, as we've talked about in relation to a few waterfall questions (see: Bonus 2022 Guide), post-filing the lender's *secured* claim is limited to the value of their collateral – and, if that collateral value has been reduced due to a transfer of (some) assets to an unsub, then that secured claim will be reduced and the lender would have an unsecured deficiency claim make up the difference.

Second, to ensure a certain level of fairness (read: that a secured creditor can't have value "stripped" from them post-filing) secured lenders are entitled to [adequate protection](#) to safeguard their collateral value (e.g., to ensure that a DIP secured by a priming lien doesn't water down their recovery from what it otherwise would be). This often comes in the form of additional liens, replacement liens, or cash payments – but only up to the *secured* value of the lender's claim which, again, could be less due to the diminished collateral base from the unsub transfer.

Third, under Section 363(k) secured creditors have the ability to [credit bid](#) in a sale process – permitting lenders to net the value of their *secured* claim off the purchase price (even if they bought their secured debt well below par pre-filing). Thus insofar as a lender's *secured* claim has fallen – because of the diminished collateral base – this valuable form of consideration also falls.

Fourth, as I touched on a bit in the Case Study Guide, if the now undersecured lender was previously oversecured, there's an added insult to injury. Because the oversecured lenders would be entitled to post-petition interest and other assorted goodies ("reasonable fees, costs, and charges") under 506(b). However, now undersecured lenders would not per se be entitled to these with their collateral value now below their total claim value.

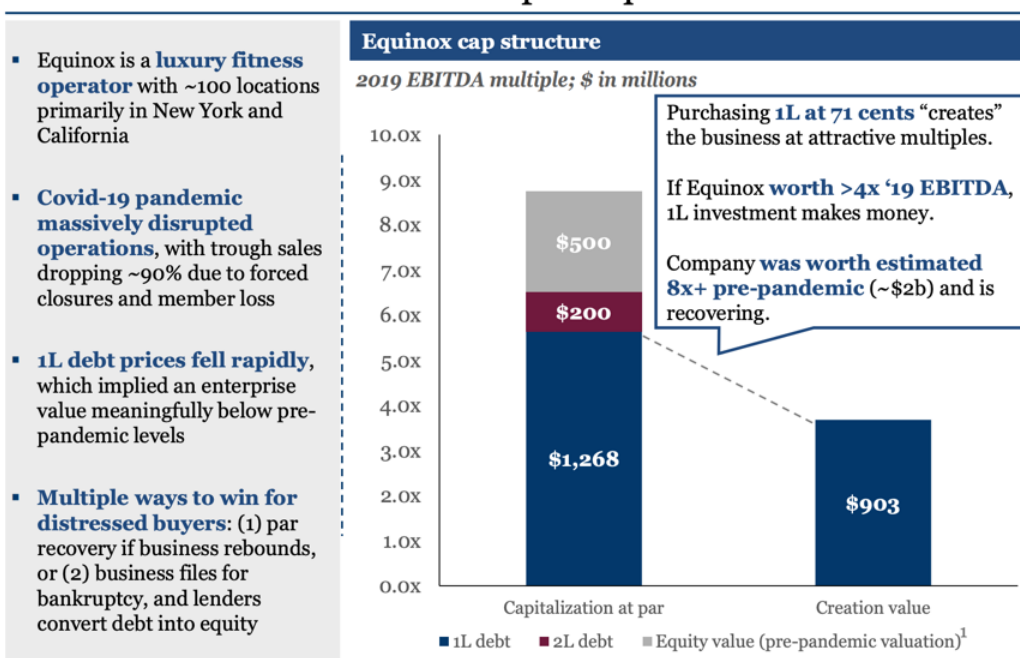


2 I've heard some people throw around the terms "creation value" and "creation multiple". What are those? Who cares about them?

Creation value is simply the market value (Face Value * Current Price/100) of the tranche you're buying into *plus* the par (face) value of all the debt (if applicable) that's higher in priority, and the creation multiple is just this creation value – typically net of cash – divided by Adj. EBITDA.

So, in a distressed situation, the creation value / multiple is the value / multiple that's most relevant to you from a recovery standpoint as it's based on your *actual* position and your *actual* cost-basis within the capital structure (read: it's assuming you're residing where value will break in the capital structure – thus why you're assuming everyone above you gets par, even if in the real world pre-filing the debt above you may be trading at levels below par as well).

Illustrative distressed debt example: Equinox



Therefore, if Redwood is buying up a company's SNs and Apollo is buying up a company's SSNs then they'll each have different creation values and creation multiples because they're "creating" the business at different levels. Further, if Redwood and Apollo are each buying the SSNs, but are doing so at different times when the SSNs are trading at different levels, then their creation values / multiples will probably be similar, but not identical, since the creation value / multiple is predicated on the actual price paid for the SSNs.

You'll often here people reference creation values / multiples by saying stuff like: "We think there's a strong possibility of this [1L / 2L / whatever] being a pull-to-par situation, but if not it's fine because we like owning the business [presuming that we get a significant amount of post-reorg equity in a future filing] at this creation value [since implicit to this creation value calculation is that we're the fulcrum and everything above us gets back par]." Or "...we had significant downside protection investing into this part of the capital structure because the creation multiple here was just 2.3x – so, even if the company files we're well covered as we think it's worth at least 5x."



Newest Investments: Swissport



Swissport is the largest independent airport services provider in the world, headquartered in Switzerland with global operations

SOURCING

- Material declines in passenger and cargo volume beginning in March 2020 due to COVID meant the company had an unsustainable cash burn
- SVP funds built a €450m position in the senior secured debt in 43 transactions over 6 months
- Directly sourced^a almost 2/3rds of senior debt position from European CLOs and other counterparties, surpassing all other distressed buyers and becoming leader of the ad hoc group

UNDERWRITING / FINANCIAL RESTRUCTURING

- SVP has been cautious in COVID-affected businesses; but Swissport is a global market-leading business with high barriers to entry, being created for ~€1.2bn v. previous acquisition EV of €3bn^b
- Drove debt-for-equity restructuring negotiations in August 2020 with senior lenders to receive >90% of the post-reorg equity upon consummation^c

OPERATIONAL VALUE ADD / STRATEGIC INITIATIVES

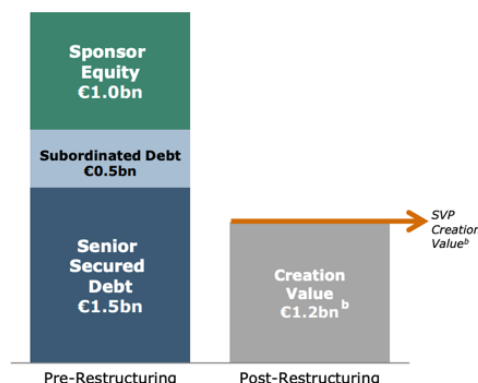
- Once restructuring is completed, SVP set to emerge as largest shareholder with 33% stake
- Identified multiple streams of post-reorg value creation with potential for an additional €150-200m EBITDA upside from operational and commercial initiatives^d
- Targeting refinancing before sale

64% SOURCED DIRECTLY^a
sourced directly from asset managers and European banks

BOUGHT INTO MARKET LEADER AT 60% DISCOUNT

IDENTIFIED €150-200M EBITDA UPSIDE^d
compared to 2019 run-rate EBITDA of €272m

AGREED DEBT FOR EQUITY RESTRUCTURING



Newest Investments: Swissport International

- Calculated as a percentage of invested capital. Sourced directly from sellers includes all transactions in which SVP believes that the interests were purchased directly from the selling institution or sourced directly from the selling institution, but transacted through a third party.
- Creation value based on average entry price for SVP funds of 78¢ for senior debt. This includes an average entry price of 72¢ of face for SVSS IV. Calculation is weighted based on invested capital and includes the average entry price of other funds and accounts managed by SVP and its affiliates.
- Equity to be received by senior lenders under agreed-upon restructuring subject to certain equity valuations.

2 Let's imagine we have a bond trading at 80 with a 10% coupon and five years to maturity. Rank the YTM's from highest to lowest if we receive coupons annually, semi-annually, or quarterly? What explains this?

In some other interview question I mentioned, as an aside, that for a bond trading below par the YTM will *decline* a touch as the frequency of coupons *increases*. Since this is perhaps a bit unintuitive, let's briefly talk about what's going on here (it wouldn't come up in an interview context but since this isn't explicitly touched on in the Bond Math Guide I figured I'd write a brief explanation).

First, try plugging in $P=80$, $FV=1000$, $Maturity=5$, $C=10\%$ into this little YTM calculator (<https://dqydj.com/bond-yield-to-maturity-calculator>) and toggle between annual, semi-annual, and quarterly frequencies. You'll see YTM's of 16.126%, 15.955%, and 15.870%, respectively.

Remember that this is all predicated on having a static bond price (\$80) regardless of coupon frequency, and the YTM is just a value that gets the PV of your cash flows equal to that static price. In the Bond Math Guide I try to provide a few different ways to think about YTM intuitively – whether you're finding total proceeds (e.g., coupons + reinvestment earnings, assuming you're reinvesting at the YTM rate) and then finding the PV of the total proceeds, or just taking the PV of your coupon + principal component (e.g., $C / (1 + YTM)^1$, $C / (1 + YTM)^2$...) and summing it up.



So, with a bond trading below par, if we think about it from a total proceeds perspective, the mechanical explanation for what's going on is that you're getting slightly more total proceeds (due to getting more coupon reinvestment income) by receiving coupons quarterly and then compounding them relative to if you're receiving coupons semi-annually (even though the quarterly YTM being used is lower). Further, you're discounting back the total proceeds at a lower YTM if you're getting coupons quarterly vs. semi-annually. *But* since there are more periods in the quarterly calculation vs. the semi-annual calculation the discount factor (denominator) ends up being a bit higher (even with the lower YTM used) so that in both cases the PV of total proceeds (whether you're getting coupons quarterly or semi-annually) equals \$80 as it should. So, the lower YTM when you have higher payment frequency is strictly necessary to make sure you get to \$80. In other words, regardless of payment frequency, if the bond price needs to equal \$80 then that's the North Star and the mathematical prerequisite of getting to that static bond price is the YTM needs to decrease as payment frequency increases. (The inverse is true when a bond is trading above par where YTM increases as payment frequency increases).

Perhaps a more intuitive (less mechanical) explanation is that if you're getting coupons sooner (e.g., quarterly vs. semi-annually) then you don't need as high of a reinvestment rate (YTM) on those coupons to get the present value of the total proceeds from holding the bond to maturity equal to the current price. Remember we found above that even with the lower quarterly YTM (reinvestment rate) the total reinvestment income earned from these coupons is a touch more than the reinvestment income earned with semi-annual coupons even though we used the slightly higher semi-annual YTM (reinvestment rate).